When Does Jump-Risk Filtering Matter for Commodity Derivatives Deep Hedging? A Semi-Synthetic Simulation Audit of Jump-Modulated No-Trade Bands, Price-Limit Locks, and Option-Gamma Spanning

Anonymous Authors

Quantitative Finance & Derivatives Microstructure Research Group

Open-Source Code & Audit Repository: https://github.com/lewygoat/Jump-Risk-Filtering-Deep-Hedging

November 8, 2026

Abstract

Strategic commodity futures such as Lithium Carbonate (`LC`) on the Guangzhou Futures Exchange (GFEX) exhibit self-exciting jump clustering alongside daily price-limit bands ($\pm 7\%$ to $\pm 15\%$) that freeze order-book execution upon limit locks. Using a protocol-specified semi-synthetic simulation audit (80 single-instrument Deep Hedging checkpoints, 60 Distributional Actor-Critic `D4PG-QR` runs, and 80 two-instrument Option-Gamma checkpoints across 100,000 S4 benchmark and 20,000 S5 stress paths), we examine when causal threshold-Hawkes intensity filtering ($\hat{\lambda}_t$) and price-limit features improve Conditional Value-at-Risk (CVaR_{95}) hedging. First, in futures-only hedging ($c = 10^{-4}$), CVaR-trained Deep Hedging (`DH1`, $\mathrm{CVaR}_{95} = 0.021078 \pm 0.000105$) reduces tail loss by 27.00% versus Black-76 delta hedging (0.028872) and by 32.64% versus mean-variance hedging (`DH_mv`, 0.031294, $p_{\mathrm{holm}} = 3.9 \times 10^{-11}$), dominating mean-variance across $\lambda_{\mathrm{MV}} \in [0.5, 50.0]$ while avoiding the 20%–30% seed collapse rate of `D4PG-QR`. Second, embedding $\hat{\lambda}_t$ into a multiplicative no-trade band (`DH_lband`, `DH_full`) yields a negative modulation slope ($\hat{\theta} = -0.0276, p = 0.0019$; $\hat{\theta} = -0.0230, p = 0.0011$) that contracts the buffer during jump bursts, cutting trading frequency by 20.11%, transaction costs by 8.09%, and the lock-step hedging gap (`LockGap`) by 29.65% on S4 and 25.67% on S5 (where `DH_full` reduces CVaR_{95} by 37.60% vs. `B76-Delta` as drift-neutralized tail risk rises +17.89%), with no statistically detectable CVaR_{95} degradation relative to `DH1` at 6× cost ($p = 0.218, p_{\mathrm{TOST}} = 0.046$). Third, Jacobian saliency ($> 99.78\%$ on core states) and oracle tests (`DH_oracle`, 0.021085) show that linear futures ($\Gamma^{\mathrm{fut}} = 0$) face a spanning ceiling. Adding a rolling 20-day ATM call option breaks this ceiling (-17.98% to -29.95%), and channeling $\hat{\lambda}_t$ through the structural no-trade band (`DH_opt_lband`, $\hat{\theta} = -0.0290, p = 1.59 \times 10^{-7}$) achieves the lowest 2D CVaR_{95} of 0.014841 ± 0.001316 (+15.41% vs. `DH_opt_cband`, $p_{\mathrm{holm}} = 0.000123$). Code is available at https://github.com/lewygoat/Jump-Risk-Filtering-Deep-Hedging.

Keywords: Deep Hedging; Hawkes Jump Process; Online Intensity Filtering; Price-Limit Microstructure; Conditional Value-at-Risk (CVaR); Multi-Instrument Gamma Hedging; Lithium Carbonate Futures

JEL Classification: G13; G11; C45; C61; Q02

1 Introduction and Related Work

1.1 Background and Research Motivation

The expansion of global green-energy supply chains has made strategic battery metals—such as Lithium Carbonate (`LC`) futures launched on the Guangzhou Futures Exchange (GFEX) in July 2023—important instruments for industrial risk management in the lithium value chain [30, 44], alongside broader clean-energy and commodity futures markets [16]. Unlike frictionless geometric Brownian diffusion [6], emerging strategic commodity futures in China exhibit two major microstructural departures documented across domestic energy, agricultural, and metal futures markets [21, 39]. First, supply-chain bottlenecks, inventory squeezes, and policy announcements generate abrupt price discontinuities [31, 36] that cluster in time, a self-exciting dynamic captured by Hawkes point processes [4, 8, 23] and marked jump-clustering specifications [14, 35]. Second, Chinese commodity futures exchanges enforce statutory daily price-limit bands ($\pm 7\%$ to $\pm 15\%$ on GFEX `LC`) accompanied by directional order-book locks when prices reach the boundary, analogous to price-limit execution freezes and volatility spillover effects examined in equity and agricultural futures markets [29, 46, 47]. During a limit lock, one-sided liquidity vanishes, freezing a hedger's ability to rebalance a short option liability when underlying Gamma convexity losses are largest.

Classical analytical hedging rules—including Black [6] practitioner delta (`B76-Delta`), discrete-time delta rebalancing [5, 7], Leland [32] volatility-adjusted delta (`Leland-Delta`), and Whalley and Wilmott [45] asymptotic no-trade bands (`WW-Band`)—rely on local diffusive Taylor expansions and continuous trade execution. As a result, their hedging performance degrades under jump clustering and execution freezes [28, 36, 43]. Introduced by Buehler et al. [10], **Deep Hedging (DH)** formulates derivative hedging as an end-to-end neural network control problem under convex risk measures such as Conditional Value-at-Risk (CVaR, Rockafellar and Uryasev 42). Subsequent work has extended Deep Hedging to long-term derivatives [12], rough volatility [26], quadratic hedging benchmarks [18], parameter ambiguity [33], equal risk pricing [13, 34], empirical equity index options [20, 37], neural no-transaction band networks [27], signature volatility models [1], backward dynamic programming with transaction costs [9, 22], and model-free distributional reinforcement learning (`D4PG-QR`, Cao et al. 11, Pickard and Lawryshyn 40).

Despite this progress, a clear research gap remains at the intersection of self-exciting jump filtering and emerging-market institutional trading frictions. Specifically, three questions remain open:

1. **When and through which architectural channel does online jump-intensity filtering matter ($\mathbf{H_0}, \mathbf{H_1}$)?** Does feeding a causal threshold-Hawkes jump-intensity proxy $\hat{\lambda}_t$ into a deep hedging policy reduce tail risk when hedging solely with linear futures, or does it require a second, convex option instrument ($\Gamma > 0$) and a structural no-trade band parameterization to extract *ex-ante* jump-Gamma protection without optimization variance?

2. **How do jump intensity and price limits reshape microstructural trading rules ($\mathbf{H_1}, \mathbf{H_3}$)?** Can a neural policy learn without sign constraints to modulate its no-trade band width $w_t(x_t)$ in response to $\hat{\lambda}_t$ and pre-position its hedge ratio ahead of price-limit locks?

3. **How do risk objectives ($\mathbf{H_2}$) and optimization methods (`BPTT` vs. `D4PG-QR`) interact with these frictions?** Does symmetric mean-variance hedging or model-free distributional reinforcement learning remain effective under self-exciting jumps and price-limit truncations?

1.2 Related Literature

Our study relates to three strands of the quantitative finance literature:

1. Deep Hedging and Convex Risk Optimization Under Market Frictions. Classical option replication under discrete rebalancing [7], jump-diffusion return discontinuities [5, 36], and proportional transaction costs [32, 45] established that market incompleteness and trading frictions prevent exact delta replication. Buehler et al. [10] established Deep Hedging via end-to-end Back-Propagation Through Time (BPTT) under proportional transaction costs and convex risk measures [42]. Carbonneau [12] and Dai et al. [18] evaluated neural networks for long-term and quadratic hedging, while Horvath et al. [26] and Lütkebohmert et al. [33] studied rough volatility and parameter ambiguity. Marzban et al. [34] and Carbonneau and Godin [13] formulated equal-risk pricing and hedging across single and multiple hedging instruments. Empirically and architecturally, Mikkilä and Kanniainen [37] and François et al. [20] showed that deep hedging outperforms practitioner delta hedging on equity index options, Na and Wan [38] and Goudenège et al. [22] developed recurrent and backward hedging schemes for American options with transaction costs, and Imaki et al. [27] and Babaei [3] analyzed no-transaction band networks and fixed-plus-proportional cost buffers. Abi Jaber and Gérard [1] and Brugière and Turinici [9] studied signature volatility models and data-augmented model-free deep hedging, while Shakeel et al. [43] examined neural SDE hedging under regime-switching jump diffusions. To the best of our knowledge, none of these studies incorporate exchange price-limit locks or identify the spanning and architectural boundary between linear futures hedging and multi-instrument Gamma hedging under Hawkes jumps.

2. Distributional Reinforcement Learning for Option Greek Hedging. Building on quantile regression distributional reinforcement learning (`QR-DQN`, Dabney et al. 17), Cao et al. [11] applied Distributional Deep Deterministic Policy Gradient (`D4PG-QR`) to hedge option Gamma and Vega under transaction costs, an approach reviewed by Pickard and Lawryshyn [40] and extended to shortfall-aware implied-volatility hedging by Chen et al. [15]. While `D4PG-QR` avoids differentiating through the market simulator, its reliance on off-policy multi-step quantile Bellman bootstrapping has not been tested against direct pathwise BPTT in finite-horizon commodity markets with self-exciting jump clusters and price-limit truncations.

3. Jump Clustering and Institutional Price Limits in Commodity Futures. Recent studies document pronounced tail risk, volatility asymmetry, and cross-market spillovers in China's lithium carbonate industry chain [30, 44], clean-energy equity–commodity portfolios [16], and Chinese crude oil futures [21, 39]. To model contagious price shocks in financial and commodity markets [2, 4], Brignone et al. [8] and Mercuri et al. [35] developed option pricing models with jump clusters and compound CARMA-Hawkes processes, while Chen et al. [14] and de Truchis et al. [19] modeled marked Hawkes jump clustering and fractional volatility persistence. Qiao et al. [41] and He et al. [25] analyzed jump components and self-exciting jumps in volatility forecasting and variance swap valuation. In addition, He et al. [24] and Jouini et al. [28] studied jump volatility forecasting and crisis-regime dynamic hedging, while Kim and Rhee [29], Zhang et al. [47], and Xia et al. [46] examined price-limit execution frictions, asymmetric risk connectedness, and session volatility in Asian equity and Chinese agricultural/energy futures markets. Combining causal online Hawkes filtering with institutional price-limit execution freezes for dynamic derivative hedging remains unexplored.

1.3 Summary of Contributions

Evaluating 80 formal 1D Deep Hedging (`DH`) checkpoints, 60 `D4PG-QR` runs, and 80 formal 2D Option-Gamma checkpoints across 10 random seeds (`9001-9010`) on 100,000 S4 benchmark paths and 20,000 S5 extreme stress paths in a controlled semi-synthetic environment partially calibrated to GFEX Lithium Carbonate (`LC`) futures, we make four calibrated contributions:

- **Contribution 1 (Asymmetric CVaR Dominance in ($\mathbf{CVaR}_{95}, \mathbf{Cost}$) over Baselines and Mean-Variance, $\mathbf{H_2}$):** Under the S4 futures-only benchmark (100,000 paths, $c = 10^{-4}$), relative to classical Black-76 delta hedging (`B76-Delta`) and symmetric mean-variance deep hedging (`DH_mv`, $\lambda_{\text{MV}} = 10$), our CVaR-trained policy (`DH1`) improves out-of-sample CVaR_{95} from 0.028872 (`B76-Delta`) and 0.031294 ± 0.000722 (`DH_mv`) to 0.021078 ± 0.000105 ($p_{\text{holm}} = 3.9 \times 10^{-11}$) while reducing transaction costs from 0.5015×10^{-3} (`DH_mv`) to 0.3166×10^{-3} and dominating `DH_mv` in the $(\text{CVaR}_{95}, \text{Cost})$ plane across $\lambda_{\text{MV}} \in [0.5, 50.0]$, at the cost of a higher loss standard deviation $\text{Std}(L)$ (0.013391 in `DH1` vs. 0.012539 in 10-seed `DH_mv` at $\lambda_{\text{MV}} = 10$ and 0.012798 at $\lambda_{\text{MV}} = 50$) because asymmetric CVaR_{95} leaves upside convex hedging gains ($L_T < 0$) unpenalized.
- **Contribution 2 (Jump-Modulated No-Trade Band and Price-Limit Pre-Positioning, $\mathbf{H_1}$ & $\mathbf{H_3}$):** Under the S4 benchmark (100,000 paths) and the S5 extreme stress regime (20,000 paths), relative to the static-band policy (`DH_cband`) and unconstrained policy (`DH1`), our Hawkes-modulated band (`DH_lband`) and full microstructure-aware policy (`DH_full`) improve S4 CVaR_{95} from 0.021512 ± 0.000133 (`DH_cband`) to 0.021187 ± 0.000183 ($p_{\text{holm}} = 0.00063$, a statistically significant microstructural adaptation with modest +1.51% 1D magnitude), reduce S4 trading frequency from 99.95% (`DH1`) to 79.85% and transaction costs from 0.3166×10^{-3} to 0.2910×10^{-3}, compress the cumulative lock-step hedging gap (`LockGap`) from 0.935×10^{-2} (`DH1`) to 0.658×10^{-2} (`DH_full`, −29.65%) on S4 and from 2.162×10^{-2} to 1.607×10^{-2} (−25.67%) on S5, and exhibit no statistically detectable CVaR_{95} degradation relative to unconstrained `DH1` under zero-shot 6× transaction costs (0.023145 vs. 0.023241, $p = 0.218$; ±1% two-one-sided equivalence $p_{\text{TOST}} = 0.046$) while saving 8.11% in friction, at the cost of a 0.52% higher CVaR_{95} on low-cost 1× S4 ($0.021187 > 0.021078$) where the 2.56×10^{-5} fee savings are smaller than the band-induced tracking error.
- **Contribution 3 (Option-Gamma Spanning and Structural Band Activation of Jump Filtering, $\mathbf{H_0}, \mathbf{H_1}, \mathbf{H_2}$):** Under the 10-seed two-instrument futures-and-rolling-ATM-option hedging environment (`TorchHedgeEnvOpt`, $\tau_h = 20$ days) on S4, relative to 1D futures policies (`DH1` 0.021078, `DH_lband` 0.021187) and 2D static-band hedging (`DH_opt_cband` 0.017546 ± 0.002298), adding the option leg reduces CVaR_{95} by 17.98%–29.95% across all 2D configurations, while embedding $\hat{\lambda}_t$ into the multiplicative no-trade band (`DH_opt_lband`, $\hat{\theta} = -0.0290 \pm 0.0064, p = 1.59 \times 10^{-7}$) achieves the lowest realistic 2D CVaR_{95} of 0.014841 ± 0.001316 (+15.41% vs. `DH_opt_cband`, $t = -7.401, p_{\text{holm}} = 0.000123$; −29.95% vs. 1D `DH_lband`), at the cost of requiring a liquid short-dated option market, whereas feeding $\hat{\lambda}_t$ into an unconstrained 2D MLP (`DH_opt1`, 0.015226 ± 0.001497) yields an 11.91% 10-seed point improvement over `DH_opt_nolam` (0.017284 ± 0.004298, $p_{\text{holm}} = 0.450$; +21.56% in the 5-seed subset) that does not achieve standalone statistical significance due to cross-seed non-convex optimization variance and information overlap with nonlinear σ_{EWMA} features (`DH_opt_ewma_nl`, 0.015904) and attenuates from +11.91% to +5.31% ($\kappa_{\text{iv}} = 2$) and +0.73% ($\kappa_{\text{iv}} = 5$) when option implied volatility co-jumps with $\hat{\lambda}_t$.
- **Contribution 4 (Optimization Stability of Direct BPTT over Distributional RL):** Under the 10-seed formal benchmark on S4 across training budgets of 3,000 to

20,000 updates, relative to model-free Distributional Actor-Critic with Quantile Regression (`D4PG-QR (C1)`), end-to-end differentiable `DH1` (BPTT) improves all-seed CVaR_{95} from 0.144200 ± 0.242296 (and non-collapsed seed mean 0.038934 ± 0.020739) to 0.021078 ± 0.000105 while reducing the seed collapse rate from 20.0%–30.0% to 0.0% (0/80 runs) and cross-seed coefficient of variation from 50.02% to 0.50% (0.09% at 20,000 updates), at the cost of requiring an explicitly known, piecewise-differentiable market transition and accounting simulator rather than purely model-free interaction.

2 Market Microstructure Model and Online Hawkes Jump Filtering

2.1 Underlying Futures Dynamics with Self-Exciting Hawkes Jumps

We consider a discrete-time commodity futures market indexed by trading days $t \in \{0, 1, \ldots, T\}$, partially calibrated to the daily settlement prices of Lithium Carbonate (`LC`) futures on the Guangzhou Futures Exchange (GFEX) [30, 44]. Let $F_t > 0$ denote the futures price at day t, normalized to $F_0 = 1.0$. Without price-limit truncations, the latent log-return $r^*_{t+1} = \ln(F^*_{t+1}/F_t)$ follows a discrete-time jump-diffusion process driven by a self-exciting Hawkes counting process [2, 4, 23]:

$$r^*_{t+1} = \left(\mu_d - \frac{1}{2}\sigma^2 - \lambda_t \bar{k}\right)\Delta t + \sigma\sqrt{\Delta t}\,\varepsilon_{t+1} + \sum_{j=1}^{N_{t+1}} J_{t+1,j}, \quad \varepsilon_{t+1} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,1), \tag{1}$$

where $\Delta t = 1$ trading day, $\sigma = 0.016144$ (0.0161 day$^{-1/2}$, 25.63% annualized) is the 5%-trimmed diffusive daily volatility estimated from 534 non-roll GFEX `LC` return observations, and $\mu_d = 0$ under the martingale futures measure with jump compensator $\bar{k} = \mathbb{E}[e^J - 1]$ [6, 36].

To capture shock clustering in energy-transition commodity markets [8, 35] and derivative variance dynamics [19, 25], the daily jump count $N_{t+1} \mid \lambda_t \sim \text{Poisson}(\lambda_t)$ obeys the discrete-time Hawkes recursion [14, 23]:

$$\lambda_t - \mu = e^{-\beta}(\lambda_{t-1} - \mu) + \alpha N_t, \quad \lambda_{-300} = \mu, \tag{2}$$

where $\mu > 0$ is the baseline intensity, $\beta > 0$ is the exponential decay rate, and $\alpha \geq 0$ is the self-excitation magnitude. In our open-source simulator (`src/simulator.py` and `src/trainer.py`, publicly available at https://github.com/lewygoat/Jump-Risk-Filtering-Deep-Hedging), each trajectory is preceded by a burn_in $= 300$-step warm-up ($300\beta = 60$ relaxation lifetimes) so that λ_0 starts from its stationary ergodic distribution rather than a deterministic cold start at μ; the simulator generates $T_{\text{sim}} = 61$ steps to span the $T = 60$-day option hedging horizon plus a one-step terminal index buffer. The stationary branching ratio and unconditional mean intensity are

$$n = \frac{\alpha}{1 - e^{-\beta}} < 1, \quad \bar{\lambda} = \mathbb{E}[\lambda_t] = \frac{\mu}{1 - n}. \tag{3}$$

In our S4 benchmark, $\bar{\lambda} = 0.04$ day^{-1}, $n = 0.3$, and $\beta = 0.20$ day^{-1} ($\mu = 0.028$, $\alpha \approx 0.05438$). Following Kou [31], jump sizes $J_{t+1,j}$ follow an asymmetric double-exponential law with upward probability $p_{\text{up}} = 0.4$ and decay rates $\eta_{\text{up}} = 40.0$ and $\eta_{\text{dn}} = 25.0$. In the GFEX `LC` sample, an event study of large daily jumps ($|r_t| > 2\sigma$, excluding locked days) reveals statistically significant post-jump continuation over horizons $h = 1, 2, 3, 4$ days, with cumulative signed returns of +0.45% (Newey-West $t = 2.09$), +0.86% ($t = 3.47$), +0.98% ($t = 2.88$), and +1.81% ($t = 2.67$). Fitting an exponential decay $\Delta C_h \propto e^{-\beta(h-1)}$ to the initial post-jump incremental returns ($h = 1, 2, 3$: +0.45%, +0.41%, +0.12%) yields $\hat{\beta}_{\text{OLS}} = 0.728$ day^{-1}, while the $h = 1$–4

continuation half-life ($t_{1/2} = \ln(2)/\beta \in [1.0, 4.0]$ trading days) implies an empirical decay range $\beta \in [0.17, 0.73]$ day^{-1} that brackets both our benchmark choice $\beta = 0.20$ day^{-1} ($t_{1/2} = 3.47$ days) and our Stage-2 fast-decay regime $\beta = 0.50$ day^{-1}. Although a finite-sample Monte Carlo audit ($\mu = 0.02, \alpha = 0.10, \beta = 0.20$) confirms that truncation-corrected MLE (`M1`) reduces limit-censoring bias in μ by 60.4% (+0.00046 vs. +0.00117 under naive MLE `M0`) and in α by 78.9% (−0.00066 vs. +0.00314), the 534-day historical sample contains only 10 tail events above 3σ, which is too small for low-variance joint point estimation of $(p_{\text{up}}, \eta_{\text{up}}, \eta_{\text{dn}}, n)$. We therefore frame our study as a **partially calibrated, semi-synthetic simulation audit** in which $(\sigma, \ell_t, p_{\text{lock}}, \beta)$ are anchored to GFEX `LC` data and $(p_{\text{up}}, \eta_{\text{up}}, \eta_{\text{dn}}, n)$ are protocol-specified scenario parameters audited across $n \in \{0.0, 0.3, 0.6\} \times \beta \in \{0.2, 0.5\}$.

2.2 Institutional Price-Limit Truncation and Execution Freeze Mechanics

Chinese commodity futures exchanges enforce daily price-limit bands (`limit_ratio` $\ell_t > 0$) relative to the prior settlement price F_t [29, 46, 47].[1] Across the 542 trading days of our GFEX `LC` calibration window, 18 statutory exchange adjustments—triggered by pre-holiday margin and price-limit widenings ahead of major Chinese public holidays (Spring Festival, National Day, and Labor Day) and delivery-month progression rules under GFEX risk-control guidelines—generated eight limit tiers: 87 days at $\ell = 7\%$ (16.05%), 116 at 8% (21.40%), 9 at 9% (1.66%), 279 at 10% (51.48%), 7 at 11% (1.29%), 3 at 12% (0.55%), 37 at 13% (6.83%), and 4 at 15% (0.74%). For each simulated path, $\ell_t \equiv \ell$ is drawn at $t = 0$ from this empirical distribution:

$$\mathbb{P}(\ell = r) = \begin{cases} 0.1605, & r = 0.07, \\ 0.2140, & r = 0.08, \\ 0.0166, & r = 0.09, \\ 0.5148, & r = 0.10, \\ 0.0129, & r = 0.11, \\ 0.0055, & r = 0.12, \\ 0.0683, & r = 0.13, \\ 0.0074, & r = 0.15. \end{cases} \tag{4}$$

When the latent price $F^*_{t+1} = F_t \exp(r^*_{t+1})$ breaches $F_t(1 \pm \ell)$, the settlement price is truncated to the boundary:

$$F_{t+1} = \text{clip}\big(F^*_{t+1},\, F_t(1-\ell),\, F_t(1+\ell)\big), \tag{5}$$

triggering $\text{limit_hit}_{t+1} = \mathbb{I}(|F^*_{t+1}/F_t - 1| \geq \ell)$. Notice an important microstructural interaction between Equations (1) and (5): while the latent jump-compensated process F^*_t is an exact martingale ($\widehat{\mathbb{E}}[F^*_T] = 1.0004 \pm 0.0005$), applying a symmetric price-limit clip $[F_t(1-\ell), F_t(1+\ell)]$ to left-skewed Kou jumps ($\eta_{\text{dn}} = 25.0 < \eta_{\text{up}} = 40.0$, $p_{\text{up}} = 0.4$, where downward jumps have a 1.6× heavier tail than upward jumps) truncates extreme downward jumps more frequently and severely than upward jumps. This asymmetric tail censoring induces a small positive cumulative drift in the observed settlement price ($\widehat{\mathbb{E}}[F_T] = 1.0029 \pm 0.0005$ over 60 steps on S4, or $\approx +4.8 \times 10^{-5}$ day^{-1}; and $\widehat{\mathbb{E}}[F_T] = 1.0162 \pm 0.0012$ on the tighter $\ell = 0.07$ S5 stress set, where our open-source `simulator.py` also provides an optional `enforce_post_clip_martingale` re-centering toggle for sensitivity verification). As we prove in Section 3, this censoring drift shifts the unconditional mean hedging loss $\mathbb{E}[L_T]$ equally for any two policies with matching average delta exposure and accounts for less than 2% of the CVaR_{95} improvement of `DH1` over `B76-Delta`.

[1] Specifically, statutory price-limit and margin adjustment rules follow the *Trading Rules of the Guangzhou Futures Exchange* and the *Risk Management Measures of the Guangzhou Futures Exchange (GFEX)*, which prescribe widened daily price-limit tiers (±7% to ±15%) around major public holidays, delivery-month progressions, and consecutive limit-hit days.

Conditional on $\text{limit_hit}_t = 1$, an order-book lock ($\text{locked_up}_t = 1$ or $\text{locked_dn}_t = 1$) occurs with probability $p_{\text{lock}} = 1/7 \approx 0.1429$, matching the historical ratio of 1 full-day lock out of 7 limit-hit days (6 limit-hit episodes totaling 38.63% cumulative absolute return; daily hit rate 1.29%, daily lock rate 0.185%). Because $p_{\text{lock}} = 1/7$ is calibrated from a small count of extreme historical events, we also audit policy sensitivity across $p_{\text{lock}} \in \{0.0, 1/14, 1/7, 1/4\}$ and partial lock-day liquidity fills $\pi_{\text{fill}} \in \{0.00, 0.10, 0.30\}$ in Section 4.2.

An exchange limit lock imposes a directional execution freeze on rebalancing. Given prior position $\delta_{t-1}^{\text{exec}}$ and target $\delta_t^{\text{target}} \in [-\delta_{\max}, \delta_{\max}]$ ($\delta_{\max} = 1.5$), an upper-limit lock ($\text{locked_up}_t = 1$) prevents long buying, while a lower-limit lock ($\text{locked_dn}_t = 1$) prevents short selling:

$$\delta_t^{\text{exec}} = \begin{cases} \min\left(\delta_t^{\text{target}}, \delta_{t-1}^{\text{exec}}\right), & \text{if } \text{locked_up}_t = 1, \\ \max\left(\delta_t^{\text{target}}, \delta_{t-1}^{\text{exec}}\right), & \text{if } \text{locked_dn}_t = 1, \\ \delta_t^{\text{target}}, & \text{otherwise.} \end{cases} \tag{6}$$

Following the exact accounting in `torch_env.py`, we measure the resulting execution friction via the cumulative pathwise unexecuted lock-step hedging gap:

$$\text{LockGap} = \mathbb{E}\left[\sum_{t=0}^{T-1} \left|\delta_t^{\text{target}} - \delta_t^{\text{exec}}\right| \mathbb{I}\big(\text{locked_up}_t \vee \text{locked_dn}_t\big)\right]. \tag{7}$$

2.3 Causal Online Threshold-Hawkes Intensity Proxy Under Price-Limit Censoring

In practice, the latent jump intensity λ_t and jump counts N_t are unobservable [4, 24, 41]. Standard threshold filters detect jumps via $|r_t| > k\sigma$, but tight price limits ($\ell = 0.07$ or 0.08) censor large jumps at the boundary. To address limit censoring while preserving causal adaptedness ($\hat{\lambda}_t \in \mathcal{F}_t$), we construct a limit-augmented causal online threshold-Hawkes intensity proxy:

$$\hat{\lambda}_t = \mu + A_t, \quad A_0 = 0, \quad A_t = e^{-\beta} A_{t-1} + \alpha\, \hat{I}_t, \tag{8}$$

where the causal jump indicator $\hat{I}_t \in \{0, 1\}$ combines a 3σ return threshold ($k = 3.0$) with the observable limit-hit indicator:

$$\hat{I}_t = \mathbb{I}\Big(|r_t| > k\sigma \ \vee\ \text{limit_hit}_t = 1\Big), \quad k = 3.0. \tag{9}$$

We audit the confusion matrix of $\hat{I}_t$ on two complementary benchmarks: (i) the unit-test diagnostic benchmark `TH2b` (30,029 true jump days, `reports/TH2b_summary.md`) and (ii) the frozen out-of-sample test set `S4_test.npz` (100,000 paths $\times 60$ steps $= 6.0 \times 10^6$ path-days containing 236,488 true jump days and 5,763,512 non-jump path-days, where 97.3% of limit-hit days correspond to true latent jumps). On `TH2b` at $k = 3.0$, the filter triggers 5,895 total detections composed of TP $= 4{,}138$ true positives and FP $= 1{,}757$ false positives (with FN $= 25{,}891$ missed moderate jumps where $|J| < 3\sigma = 0.0484$), yielding precision TP/(TP + FP) $= 70.20\%$, true-positive recall TP/(TP + FN) $= 13.78\%$ (whereas the ratio (TP + FP)/(TP + FN) $= 5{,}895/30{,}029 = 19.63\%$ represents the raw detection-to-true-jump count ratio), linear correlation $r(\hat{\lambda}_t, \lambda_t) = 0.4388$, and mean filtered intensity $\mathbb{E}[\hat{\lambda}_t] = 0.02465$ versus $\mathbb{E}[\lambda_t] = 0.04496$ (relative bias -45.17%). On the frozen `S4_test.npz` benchmark at $k = 3.0$, the filter triggers 48,883 detections (TP $= 44{,}052$, FP $= 4{,}831$, FN $= 192{,}436$), achieving precision 90.12%, recall 18.63%, linear correlation $r = 0.4455$, and $\mathbb{E}[\hat{\lambda}_t] = 0.03019$ versus $\mathbb{E}[\lambda_t] = 0.03995$ (relative bias -24.44%). Varying the threshold across $k \in \{2.0, 2.5, 3.0, 3.5\}$ on `S4_test.npz` reveals a standard precision–recall trade-off: lowering k to 2.0 or 2.5 raises recall to 33.40% (TP $= 78{,}995$, FP $= 70{,}626$, precision 52.80%, $r = 0.4555$, bias -13.17%)

and 24.80% (TP $= 58,638$, FP $= 13,704$, precision 81.06%, $r = 0.4853$, bias -21.81%) at the expense of frequent diffusive false alarms, whereas raising k to 3.5 increases precision to 91.50% (TP $= 33,535$, FP $= 3,114$) but lowers recall to 14.18% ($r = 0.3947$, bias -25.81%).[2] In our multiplicative no-trade band architecture $w_t(x_t) = \mathrm{softplus}(g_\phi(x_{t,-\lambda}))\exp(\theta\hat{\lambda}_t)$ (Section 3), any constant scale underestimation $\hat{\lambda}_t \approx c_\lambda \lambda_t$ ($c_\lambda \in [0.55, 0.76]$) is automatically absorbed by the unconstrained learnable slope θ via $\tilde{\theta} = \theta/c_\lambda$ and the intercept of g_ϕ. Evaluating both $\hat{\lambda}_t$ (`DH1`) and true λ_t (`DH_oracle`) separates residual filter noise from instrument-spanning limits.

3 Microstructure-Aware Deep Hedging Architecture

3.1 Liability Valuation, Wealth Accounting, and Multi-Instrument Extension

We consider a financial intermediary hedging a short European at-the-money (ATM) call option on the futures contract with strike $K = F_0 = 1.0$ and maturity $T = 60$ trading steps ($t = 0, \ldots, 60$). Following the valuation routines in `src/pricing76.py`, at $t = 0$ the option premium $P_0 = \widehat{\mathbb{E}}[\max(F_T - K, 0)]$ is priced at the empirical martingale expectation of the terminal payoff across the simulated ensemble, and the constant Black-76 implied volatility $\sigma_{\mathrm{imp}} = C_{\mathrm{B76}}^{-1}(F_0, K, T, P_0)$ is inverted from P_0 [6]. On the S4 benchmark (100,000 paths), this yields $P_0 = 0.058595$ and $\sigma_{\mathrm{imp}} = 0.018822$ day$^{-1/2}$ (29.88% annualized); on the S5 extreme stress set (20,000 paths), $P_0 = 0.064603$ and $\sigma_{\mathrm{imp}} = 0.020756$ day$^{-1/2}$ (32.95% annualized). Why does $\sigma_{\mathrm{imp}} = 0.018822$ exceed the diffusive volatility $\sigma = 0.016144$ by +16.59% on S4? Under the jump-diffusion process in Equation (1), total daily return variance is $\mathrm{Var}(r_t) \approx \sigma^2 + \bar{\lambda}\mathbb{E}[J^2]$; because P_0 equals the jump-inclusive martingale option value, inverting the Black-76 formula at P_0 maps the expected jump-variance contribution $\bar{\lambda}\mathbb{E}[J^2]$ into σ_{imp}. Consequently, $\Delta_{\mathrm{B76},t}(\sigma_{\mathrm{imp}})$ and $\Gamma_{\mathrm{B76},t}(\sigma_{\mathrm{imp}})$ represent practitioner jump-inclusive implied Greeks rather than pure diffusive Greeks. For $t < T$, the liability is marked to market at $V_t = C_{\mathrm{B76}}(F_t, K, T - t, \sigma_{\mathrm{imp}})$, settling at exact payoff $V_T = \max(F_T - K, 0)$ [13, 34].

1. Single-Instrument Futures Hedging (`TorchHedgeEnv`). With one-way proportional transaction cost rate $c = 10^{-4}$, cumulative trading cost C_t^{fut} and futures gain G_t^{fut} evolve from $\delta_{-1}^{\mathrm{exec}} = 0$ as

$$C_t^{\mathrm{fut}} = \sum_{s=0}^{t-1} cF_s \left|\delta_s^{\mathrm{exec}} - \delta_{s-1}^{\mathrm{exec}}\right| + \mathbb{I}(t = T)\, cF_T \left|\delta_{T-1}^{\mathrm{exec}}\right|, \quad G_t^{\mathrm{fut}} = \sum_{s=0}^{t-1} \delta_s^{\mathrm{exec}}(F_{s+1} - F_s). \tag{10}$$

Net portfolio wealth is $W_t = P_0 - V_t + G_t^{\mathrm{fut}} - C_t^{\mathrm{fut}}$, and normalized terminal hedging loss is $L_T = -W_T/F_0$. Because $P_0 = \widehat{\mathbb{E}}[V_T]$ by construction, taking the unconditional expectation of L_T cancels the option premium and terminal payoff identically across all hedging policies:

$$\mathbb{E}[L_T] = \frac{\mathbb{E}[V_T] - P_0}{F_0} - \frac{\mathbb{E}[G_T^{\mathrm{fut}}]}{F_0} + \frac{\mathbb{E}[C_T^{\mathrm{fut}}]}{F_0} = -\frac{\mathbb{E}[G_T^{\mathrm{fut}}]}{F_0} + \frac{\mathbb{E}[C_T^{\mathrm{fut}}]}{F_0}. \tag{11}$$

Consequently, when asymmetric price-limit clipping of left-skewed jumps induces a small positive drift $\mathbb{E}[F_{s+1} - F_s] > 0$ (Section 2), any long delta hedge ($\delta_s^{\mathrm{exec}} > 0$) earns a small positive expected futures gain $\mathbb{E}[G_T^{\mathrm{fut}}] \approx \bar{\delta}\,\mathbb{E}[F_T - F_0] > \mathbb{E}[C_T^{\mathrm{fut}}]$, explaining why $\mathrm{Mean}(L) \approx -0.0023$ on

[2] Note that in the `S4_test.npz` threshold sweep, $|r_t|$ is normalized by the effective untrimmed return scale $\sigma_{\mathrm{eff}} \in [0.0180, 0.0202]$ (matching the raw empirical return SD $\sigma_{\mathrm{raw}} = 0.02038$ in Section 4.1) together with price-limit boundary clipping (limit_hit$_t = 1$ at $\ell = 0.07$), so the realized non-jump false-positive rates (FP$/5.7635 \times 10^6 = 1.225\%, 0.238\%, 0.084\%, 0.054\%$ across $k \in \{2.0, 2.5, 3.0, 3.5\}$) differ from static textbook $\mathcal{N}(0,1)$ tail probabilities.

S4 and -0.0072 on S5. Crucially, because `DH1` and `B76-Delta` maintain nearly identical average delta exposure ($\mathrm{MAE}(\delta^{\mathrm{DH1}}, \delta^{\mathrm{B76}}) < 0.03$, and 0.00523 across neural policies in Table 2), differential drift capture between `DH1` and `B76-Delta` is bounded by $|\Delta\bar{\delta}| \cdot |\mathbb{E}[F_T - F_0]| < 1.5 \times 10^{-4}$ ($< 2\%$ of the 0.007794 CVaR_{95} reduction). Furthermore, to compare tail risk across regimes with different limit-censoring drifts (such as S4 vs. S5), we also report the **drift-neutralized (de-meaned) tail risk** $\widetilde{\mathrm{CVaR}}_{95}(L_T) \equiv \mathrm{CVaR}_{95}(L_T - \mathbb{E}[L_T]) = \mathrm{CVaR}_{95}(L_T) - \mathrm{Mean}(L)$ by translation invariance [42].

2. Multi-Instrument Option-and-Futures Gamma Hedging (`TorchHedgeEnvOpt`). To test whether jump filtering requires a convex instrument to span jump-Gamma risk [5, 13, 43], `TorchHedgeEnvOpt` (`src/torch_env_opt.py`) adds a rolling short-dated ATM call option with time-to-maturity $\tau_h = 20.0$ days (`TAU_H` $= 20.0$) and position bound $\eta_t^{\mathrm{target}} \in [-\eta_{\max}, \eta_{\max}]$ ($\eta_{\max} = 2.0$). At day t, a unit ATM call ($K_t^{\mathrm{roll}} = F_t$) opens at $V_{\mathrm{open},t} = C_{\mathrm{B76}}(F_t, F_t, \tau_h, \sigma_{\mathrm{imp}})$ (with sample mean $\overline{V}_{\mathrm{open}} = 0.033311 \approx 3.33\%$ of F_0 on S4) and closes at day $t+1$ at $V_{\mathrm{close},t+1} = C_{\mathrm{B76}}(F_{t+1}, F_t, \tau_h - 1, \sigma_{\mathrm{imp}})$, yielding one-step option PnL $\Delta V_t^{\mathrm{opt}} = V_{\mathrm{close},t+1} - V_{\mathrm{open},t}$ and Black-76 Gamma $\Gamma_{\mathrm{B76},t} = \phi(d_{1,t})/(F_t \sigma_{\mathrm{imp}} \sqrt{\tau_h})$. When the futures contract is limit-locked ($\mathrm{locked_up}_t \vee \mathrm{locked_dn}_t = 1$), option rebalancing is frozen ($\eta_t^{\mathrm{exec}} = \eta_{t-1}^{\mathrm{exec}}$). In the baseline controlled experiment, the option leg incurs proportional cost rate $c_{\mathrm{opt}} = c = 10^{-4}$ applied to the option premium $V_{\mathrm{open},t}$ (so the per-contract dollar cost is $c_{\mathrm{opt}} V_{\mathrm{open},t} \approx 0.033 c F_t$), with total two-instrument cumulative cost and option gain given by

$$C_T^{\mathrm{tot}} = C_T^{\mathrm{fut}} + \sum_{s=0}^{T-1} c_{\mathrm{opt}} V_{\mathrm{open},s} \left|\eta_s^{\mathrm{exec}} - \eta_{s-1}^{\mathrm{exec}}\right| + c_{\mathrm{opt}} V_{\mathrm{open},T} \left|\eta_{T-1}^{\mathrm{exec}}\right|, \quad G_T^{\mathrm{opt}} = \sum_{s=0}^{T-1} \eta_s^{\mathrm{exec}} \Delta V_s^{\mathrm{opt}}. \tag{12}$$

Because constant σ_{imp} and $c_{\mathrm{opt}} = c$ isolate pure Gamma spanning from implied-volatility risk premiums and wider option bid-ask spreads, Section 4.4 explicitly audits policy robustness under option cost multipliers $c_{\mathrm{opt}}/c \in \{1\times, 5\times, 10\times\}$ and jump-driven dynamic implied volatility $\sigma_{\mathrm{imp},t} = \sigma_{\mathrm{imp},0}(1 + \kappa_{\mathrm{iv}} \hat{\lambda}_t)$ for $\kappa_{\mathrm{iv}} \in \{0, 2, 5\}$.

3.2 Information Sets, Policy Network, and Multiplicative Intensity-Modulated No-Trade Band

To isolate each microstructural component across Hypotheses $\mathbf{H_0}, \mathbf{H_1}, \mathbf{H_2}, \mathbf{H_3}$, we define nested causal state vectors $x_t \in \mathbb{R}^d$ [22, 38]:

1. **Core State Features (6 dimensions):** normalized time-to-maturity $\tau_t = (T-t)/T$, log-moneyness $\ln(F_t/K)$, prior executed hedge $\delta_{t-1}^{\mathrm{exec}}$, normalized wealth W_t/F_0, Black-76 delta $\Delta_{\mathrm{B76},t}$, and EWMA volatility $\sigma_{\mathrm{EWMA},t}$ ($\sigma_0 = 0.0161$, $v_t = 0.94 v_{t-1} + 0.06 r_t^2$).

2. **Baseline Limit Ratio (1 dimension):** $\ln(1 + \ell_t)$, included in all baseline states (`DH_nolam`, $d = 7$).

3. **Jump-Intensity Signal (1 dimension, $\mathbf{H_0}$):** filtered intensity $\hat{\lambda}_t$ (`DH1`, $d = 8$) or true intensity λ_t (`DH_oracle`, $d = 8$).

4. **Explicit Price-Limit Features (3 dimensions, $\mathbf{H_3}$):** lock indicator $\mathbb{I}(\mathrm{locked_up}_t \vee \mathrm{locked_dn}_t)$, consecutive locked days $\mathrm{consec_lock_days}_t$, and raw limit ratio ℓ_t (`DH_limit` and `DH_full`, $d = 11$).

5. **Two-Instrument Option Features (2 dimensions in `TorchHedgeEnvOpt`):** prior option position $\eta_{t-1}^{\mathrm{exec}}$ and rolling Gamma $\Gamma_{\mathrm{B76},t}$.

For direct policies without a no-trade band (`DH_nolam`, `DH1`, `DH_oracle`, `DH_limit`, `DH_mv`), a $[128, 128]$ ReLU MLP $f_\psi : \mathbb{R}^d \to \mathbb{R}$ maps x_t to the target hedge position:

$$\delta_t^{\text{target}} = \delta_{\max} \tanh\big(f_\psi(x_t)\big), \quad \delta_{\max} = 1.5. \tag{13}$$

To test Hypothesis $\mathbf{H_1}$ [27, 45], we introduce a multiplicative no-trade band architecture (`BandPolicy`, extending neural band networks Babaei 3, Imaki et al. 27). A $[128, 128]$ ReLU center network $f_\psi(x_t)$ outputs $c_t = \delta_{\max} \tanh(f_\psi(x_t))$, while a $[32, 32]$ ReLU band network $g_\phi(x_{t,-\lambda})$ (taking all state features *excluding* $\hat{\lambda}_t$, with output weights initialized to 0.0 and bias to -3.0) sets the baseline half-width. In `DH_cband`, $w_t(x_t) = \text{softplus}(g_\phi(x_{t,-\lambda}))$. In `DH_lband` and `DH_full`, the half-width is modulated by $\hat{\lambda}_t$ via an unconstrained learnable scalar $\theta \in \mathbb{R}$ (initialized at $\theta_0 = 0.0$):

$$w_t(x_t) = \text{softplus}\big(g_\phi(x_{t,-\lambda})\big) \cdot \exp\big(\theta\, \hat{\lambda}_t\big). \tag{14}$$

The target position projects $\delta_{t-1}^{\text{exec}}$ onto $[c_t - w_t(x_t),\, c_t + w_t(x_t)]$ via the piecewise-differentiable soft-projection operator:

$$\delta_t^{\text{target}} = \text{clip}\Big(\delta_{t-1}^{\text{exec}} + \text{soft_proj}\big(\delta_{t-1}^{\text{exec}} - c_t,\, w_t(x_t)\big),\, -\delta_{\max},\, \delta_{\max}\Big), \tag{15}$$

where $\text{soft_proj}(u, w) = -\text{ReLU}(u - w) + \text{ReLU}(-u - w)$. When $\theta < 0$, higher jump intensity $\hat{\lambda}_t$ contracts $w_t(x_t)$ to enforce tighter rebalancing during jump clusters, whereas lower $\hat{\lambda}_t$ widens $w_t(x_t)$ to save transaction costs. Testing whether unconstrained training yields $\hat{\theta} < 0$ across random seeds provides a direct test of microstructural adaptation.

3.3 Risk Objectives and End-to-End BPTT vs. Distributional RL Optimization

Following Buehler et al. [10] and Rockafellar and Uryasev [42], our primary objective minimizes the Conditional Value-at-Risk (CVaR_α, $\alpha = 0.95$) of L_T [9, 12]:

$$\min_{\Theta,\, b \in \mathbb{R}} \mathcal{J}_{\text{CVaR}}(\Theta, b) = b + \frac{1}{(1-\alpha)B} \sum_{i=1}^{B} \max\big(L_T^{(i)}(\Theta) - b,\, 0\big), \tag{16}$$

where Θ denotes policy parameters, $b \in \mathbb{R}$ is the jointly learned VaR_α threshold, and $B = 1024$ is the batch size. To test Hypothesis $\mathbf{H_2}$, we compare $\mathcal{J}_{\text{CVaR}}$ against the symmetric mean-variance objective (`DH_mv`, Dai et al. 18):

$$\min_{\Theta} \mathcal{J}_{\text{MV}}(\Theta; \lambda_{\text{MV}}) = \widehat{\mathbb{E}}\big[L_T(\Theta)\big] + \frac{\lambda_{\text{MV}}}{2} \widehat{\text{Var}}\big(L_T(\Theta)\big), \tag{17}$$

with $\lambda_{\text{MV}} = 10.0$ in the main benchmark and $\lambda_{\text{MV}} \in \{0.5, 1.0, 2.0, 5.0, 10.0, 25.0, 50.0\}$ in our frontier sweep.

Because the accounting and price-limit rules are piecewise differentiable almost everywhere, our `DH` policies optimize $\mathcal{J}_{\text{CVaR}}$ via end-to-end Back-Propagation Through Time (BPTT) using Adam (`actor_lr` $= 3 \times 10^{-4}$, gradient clip 5.0). To compare BPTT against the model-free distributional reinforcement learning approach of Cao et al. [11] and Pickard and Lawryshyn [40] (see also Chen et al. 15), we implement a Distributional Deep Deterministic Policy Gradient with Quantile Regression (`D4PG-QR`, `C1-C9`) critic with $M_q = 51$ quantile atoms trained via quantile Huber loss ($\kappa_h = 1.0$, Dabney et al. 17) on one-step Rockafellar-Uryasev rewards $r_t = -\big((L_t - b)^+ - (L_{t-1} - b)^+\big)$ ($\tau_{\text{polyak}} = 0.005$, buffer $100,000$, batch 256).

4 Empirical Results and Microstructural Diagnostics

4.1 Calibration Data, Benchmark Policies, and Statistical Protocol

The diffusive volatility, price-limit tier distribution, conditional lock probability, and post-jump continuation half-life are calibrated to historical daily settlement prices of the generic front-month Lithium Carbonate (`LC`) futures contract on the Guangzhou Futures Exchange (GFEX) [30, 44] over the window `2023-07-24` to `2025-10-20` (542 trading days, yielding 534 non-roll daily log-returns). In this calibration sample, raw daily log-returns exhibit a standard deviation of 0.02038, excess kurtosis of 2.078, skewness of 0.113, a 99th-percentile absolute return of 0.0624, a median-absolute-deviation volatility estimate of 0.01465, and a 5%-trimmed diffusive volatility of $\sigma = 0.016144$ (0.0161 day$^{-1/2}$).

To prevent data snooping and selection bias [20, 37], our experimental pipeline (archived in `configs/experiment_protocol.json` at https://github.com/lewygoat/Jump-Risk-Filtering-Deep-Hedging) enforces a strict separation between hyperparameter tuning, model training, checkpoint selection, and out-of-sample testing:

1. **Training, Validation, and Frozen Test Sets:** Learning rates (`actor_lr` $= 3 \times 10^{-4}$, `critic_lr` $= 10^{-3}$ searched over $\{10^{-4}, 3\times10^{-4}, 10^{-3}\}$), batch size (1024 paths for `DH`, 256 transitions for `D4PG-QR`), and architecture widths ($[128, 128]$) are locked using three independent tuning seeds (`8001-8003`). Formal policies are trained in PyTorch 2.x (`float64` precision) on a 64-bit Windows workstation for $8,000$ gradient updates (with cross-budget sensitivity evaluated at $3,000$, $10,000$, and $20,000$ updates) and evaluated every 500 updates on a fixed validation set (`val_set.npz`, $10,000$ paths). For every run, the checkpoint achieving the lowest validation CVaR_{95} is frozen as `best.pt`.

2. **Benchmark (S4) and Out-of-Distribution Stress (S5) Markets:** All frozen checkpoints are evaluated once on the primary benchmark test set **S4** (`S4_test.npz`, $N = 100,000$ paths, $T = 60$ hedging steps; $n = 0.3, \beta = 0.2, \bar{\lambda} = 0.04$, empirical limit distribution; $P_0 = 0.058595$, $\sigma_{\text{imp}} = 0.018822$ day$^{-1/2}$; pathwise jump incidence 83.03% [$83,028$ paths], limit-hit path incidence 16.51% [$16,506$ paths], locked-path incidence 3.00% [$2,997$ paths]) and zero-shot transferred without retraining to the out-of-distribution (OOD) crisis stress set **S5** (`S5_test.npz`, $N = 20,000$ paths; branching ratio doubled to $n = 0.6$ at fixed $\bar{\lambda} = 0.04$, tight uniform price limit $\ell = 0.07$, jump amplitude scale $\times 1.5$; $P_0 = 0.064603$, $\sigma_{\text{imp}} = 0.020756$ day$^{-1/2}$; pathwise jump incidence 65.80% [$13,159$ paths], limit-hit path incidence 35.77% [$7,153$ paths], locked-path incidence 10.08% [$2,016$ paths]) [21, 28]. Note that in S5, doubling n from 0.3 to 0.6 at fixed $\bar{\lambda} = 0.04$ reduces the baseline immigration rate from $\mu = 0.028$ to $\mu = 0.016$ day^{-1} ($\mathbb{P}(N_{1:60} = 0) \approx e^{-60\mu} \approx 38.3\%$), lowering the fraction of paths with at least one jump from 83.03% to 65.80% while concentrating jumps into violent self-exciting clusters that more than double the limit-hit path incidence (35.77% vs. 16.51%) and triple the locked-path incidence (10.08% vs. 3.00%). To audit tail performance on paths where price limits bind, we report `Cond. CVaR95 (Hit-Sub)` and `Cond. CVaR95 (Lock-Sub)`, defined as the conditional 95% CVaR (mean of the worst 5% terminal losses) evaluated *within* the subset of paths experiencing at least one limit hit ($\mathcal{S}_{\text{hit}}$: $N = 16,506$ on S4, top-5% tail of 825 paths; $N = 7,153$ on S5, top-5% tail of 358 paths) or at least one limit lock ($\mathcal{S}_{\text{lock}}$: $N = 2,997$ on S4, top-5% tail of 150 paths; $N = 2,016$ on S5, top-5% tail of 101 paths), respectively.

3. **Classical Analytical and Heuristic Baselines:** We benchmark neural policies against four non-neural strategies: (i) `No-Hedge` ($\delta_t \equiv 0$); (ii) `B76-Delta` practitioner delta $\delta_t^{\text{target}} = \Phi(d_{1,t})$ [6, 7]; (iii) `Leland-Delta` transaction-cost-adjusted volatility delta using modified variance $\tilde{\sigma}^2 = \sigma_{\text{imp}}^2\big(1 + \sqrt{2/\pi} \cdot \frac{2c}{\sigma_{\text{imp}}\sqrt{\Delta t}}\big)$ [32]; and (iv) `WW-Band` asymptotic

no-trade band around $\Delta_{\mathrm{B76},t}$ with half-width $w_t^{\mathrm{WW}} = \left(\frac{3cF_t\Gamma^2_{\mathrm{B76},t}}{2\gamma_{\mathrm{WW}}}\right)^{1/3}$ calibrated at risk aversion $\gamma_{\mathrm{WW}} = 0.25$ and $\gamma_{\mathrm{WW}} = 1.0$ [45].

4. **Multi-Seed Statistical Testing, Path-Sampling Uncertainty, and Minimum Detectable Effect (MDE):** Each neural specification in both the 1D and 2D benchmarks is trained across $M = 10$ independent random seeds (`9001-9010`) using unbiased sample standard deviations (`ddof=1`); in addition to cross-seed SD, path-block bootstrap standard errors across the $N = 100,000$ S4 test paths ($\mathrm{SE}_{\mathrm{path}} \approx 0.00018$) and $N = 20,000$ S5 test paths ($\mathrm{SE}_{\mathrm{path}} \approx 0.00042$) confirm that path-sampling noise is an order of magnitude smaller than the primary treatment effects. Across our $m = 4$ protocol-specified hypothesis family $\mathcal{H} = \{\mathbf{H_0}$ (`DH1` vs. `DH_nolam`), $\mathbf{H_1}$ (`DH_lband` vs. `DH_cband`), $\mathbf{H_2}$ (`DH1` vs. `DH_mv`), $\mathbf{H_3}$ (`DH_limit` vs. `DH1`)$\}$, we report cross-seed Mean $\pm$ SD, paired two-sided Student's t-tests, Wilcoxon signed-rank tests, 10,000-resample paired bootstrap 95% confidence intervals (CI), and Holm-Bonferroni step-down adjusted p-values (p_{holm}). In the 2D benchmark, all formal statistical inferences are based on the complete $M = 10$ seed sample, while we also report a post-hoc $n = 9$ sensitivity check excluding seed `9003`, which satisfies the standard $> 3 \times$ IQR local-minimum outlier criterion in `DH_opt_nolam` (0.029447) and `DH_opt_limit` (0.027975). We distinguish between our protocol-specified 10% practical economic relevance threshold and the statistical minimum detectable relative effect size at two-sided $\alpha = 0.05$ and 80% power, $\mathrm{MDE}_{80\%} = (t_{0.975,M-1} + t_{0.80,M-1}) \cdot \mathrm{SD}(\Delta)/(\sqrt{M}\,\overline{\mathrm{CVaR}}_{\mathrm{base}})$, where $t_{0.975,9} + t_{0.80,9} = 2.262 + 0.883 = 3.145$ for $M = 10$ (4.110 for $M = 5$). In the 1D S4 benchmark ($M = 10$), tight cross-seed pairing yields $\mathrm{MDE}_{80\%} \in [0.45\%, 1.80\%]$; in the 2D option-hedging benchmark ($M = 10$), $\mathrm{MDE}_{80\%}$ is 6.55% for $\mathbf{H_1}$ (`DH_opt_lband` vs. `DH_opt_cband`, 4.18% on $n = 9$ converged seeds) and 5.46% for $\mathbf{H_2}$ (`DH_opt1` vs. `DH_opt_mv`), but 28.76% for unconstrained $\mathbf{H_0}$ (`DH_opt1` vs. `DH_opt_nolam`) due to higher optimization variance in unbanded 2D networks. Under our protocol-specified decision rule, a hypothesis is declared both statistically and economically activated only if $p_{\mathrm{holm}} < 0.05$ and its relative CVaR_{95} reduction exceeds both 10% and $\mathrm{MDE}_{80\%}$; under our protocol-specified title-framing contingency rule, because unconstrained 2D jump filtering ($\mathbf{H_0}$, `DH_opt1` vs. `DH_opt_nolam`) yields +11.91% ($p_{\mathrm{holm}} = 0.450 > 0.05$) across 10 seeds whereas structural band modulation ($\mathbf{H_1}$, `DH_opt_lband` vs. `DH_opt_cband`) achieves +15.41% ($p_{\mathrm{holm}} = 0.000123 < 0.05$), the manuscript title and thesis are framed around conditional mechanism inquiry (*"When Does Jump-Risk Filtering Matter..."*) rather than unconditional unconstrained jump-filtering superiority.

4.2 Baseline Performance on S4 and Microstructural Validation of $\mathbf{H_1}$ and $\mathbf{H_2}$

Table 1 summarizes the out-of-sample hedging performance of all classical baselines and neural Deep Hedging specifications across the 100,000 paths of the S4 benchmark market ($c = 10^{-4}$). Consistent across all 10 random seeds, every CVaR-trained Deep Hedging policy reduces tail risk relative to the analytical benchmarks: `DH1` achieves an out-of-sample CVaR_{95} of 0.021078 ± 0.000105, representing a 27.00% reduction relative to `B76-Delta` (0.028872), a 26.88% reduction relative to `Leland-Delta` (0.028826), and a 29.37% reduction relative to `WW-Band` ($\gamma_{\mathrm{WW}} = 0.25$, 0.029841).

Three primary empirical conclusions emerge from Table 1 and Figure 1:

1. **Statistical Confirmation of $\mathbf{H_2}$ (Asymmetric CVaR vs. Symmetric Mean-Variance):** Optimizing the Rockafellar-Uryasev CVaR_{95} objective (`DH1`) reduces out-of-sample CVaR_{95} by 32.64% relative to the mean-variance policy `DH_mv` (0.021078

Table 1: Out-of-sample hedging performance on the S4 benchmark (100,000 paths, 10 random seeds `9001-9010`, $c = 10^{-4}$). Bold entries highlight best or key microstructural metrics.

Policy / Configuration	CVaR_{95} (Mean ± SD)	VaR_{95}	CVaR_{99}	Std(L)	Cost ($\times 10^{-3}$)	Turnover	Trade Freq.	Lock Gap ($\times 10^{-2}$)	Cond. CVaR_{95} (Hit-Sub)	Cond. CVaR_{95} (Lock-Sub)
`No-Hedge`	0.272131	0.195324	0.386982	0.091207	0.0000	0.0000	0.00%	0.000	0.294908	0.280819
`B76-Delta`	0.028872	0.018049	0.047693	0.010983	0.3344	3.2907	97.42%	0.677	0.049075	0.069395
`Leland-Delta`	0.028826	0.018039	0.047602	0.010991	0.3340	3.2860	97.45%	0.676	0.048992	0.069287
`WW-Band` ($\gamma_{WW} = 0.25$)	0.029841	0.018857	0.048804	0.011688	0.2400	2.3456	62.72%	0.623	0.050307	0.070028
`WW-Band` ($\gamma_{WW} = 1.0$)	0.036137	0.023888	0.057094	0.015920	0.1690	1.6315	34.85%	0.450	0.058638	0.076012
`DH_nolam` (No $\hat{\lambda}_t$)	0.021073 ± 0.000166	0.013791	0.036855	0.013483	0.3137	3.0830	99.95%	0.931	0.037057	0.059718
`DH1` (Filtered $\hat{\lambda}_t$)	0.021078 ± 0.000105	0.013779	0.036973	0.013391	0.3166	3.1123	99.95%	0.935	0.037077	0.059416
`DH_oracle` (True λ_t)	0.021085 ± 0.000164	0.013734	0.036963	0.013313	0.3152	3.0981	99.95%	0.934	0.037069	0.059465
`DH_cband` (Const. Band)	0.021512 ± 0.000133	0.014144	0.037043	0.013261	0.2650	2.5995	58.05%	0.926	0.037152	0.058799
`DH_lband` ($\hat{\lambda}_t$-**Band**)	$\mathbf{0.021187 \pm 0.000183}$	0.014002	**0.036610**	0.013468	**0.2910**	**2.8575**	**79.85%**	0.912	**0.036848**	**0.058691**
`DH_limit` **(Limit State)**	$\mathbf{0.021021 \pm 0.000088}$	**0.013653**	0.037003	0.013298	0.3165	3.1097	99.95%	1.113	0.037213	0.059824
`DH_full` **(Full Model)**	0.021131 ± 0.000190	0.013840	0.036892	0.013326	**0.2986**	**2.9312**	**80.76%**	**0.658**	0.037075	0.059492
`DH_mv` (Mean-Variance)	0.031294 ± 0.000722	0.018533	0.054044	**0.012539**	0.5015	5.0256	99.95%	1.405	0.051650	0.075797

Notes: For all neural policies, CVaR_{95} is reported as cross-seed Mean ± SD across $M = 10$ independent training seeds (`9001-9010`, sample standard deviation `ddof=1`); deterministic analytical baselines have SD = 0 (path-block bootstrap $\text{SE}_{\text{path}} \approx 0.00018$ across $N = 100,000$ paths). Cross-seed SD in this 1D futures table is of order 10^{-4} (0.000088–0.000722), whereas in the 2D option-hedging table (Table 3) cross-seed SD is of order 10^{-3} due to joint optimization over (δ_t, η_t). Cost ($C_T^{\text{fut}}/F_0 \times 10^{-3}$) and Turnover ($\sum_{t=0}^{T-1} |\delta_t^{\text{exec}} - \delta_{t-1}^{\text{exec}}| + |\delta_{T-1}^{\text{exec}}|$) include terminal liquidation at $t = T$. `Cond. CVaR95 (Hit-Sub)` and `Cond. CVaR95 (Lock-Sub)` denote the conditional 95% CVaR computed over the worst 5% terminal losses *within* the subset of paths experiencing at least one price-limit hit ($N = 16,506$, tail size 825) or at least one limit lock ($N = 2,997$, tail size 150), respectively. For `DH1` and `DH_full`, unconditional Mean(L) is −0.002306 and −0.002286, yielding drift-neutralized tail risk $\widetilde{\text{CVaR}}_{95} = \text{CVaR}_{95} - \text{Mean}(L)$ of 0.023384 and 0.023417, respectively.

vs. 0.031294; paired difference $\Delta = -0.010216$, 95% bootstrap CI $[-0.010644, -0.009760]$, $t = -43.091$, $p = 9.7 \times 10^{-12}$, $p_{\text{holm}} = 3.9 \times 10^{-11}$, Wilcoxon $p = 0.00195$). In addition, `DH_mv` incurs 58.40% higher transaction costs (0.5015×10^{-3} vs. 0.3166×10^{-3}) and a turnover of 5.0256 because quadratic variance penalizes upside hedging windfalls ($L_T < 0$) and downside tail losses ($L_T > 0$) symmetrically, achieving a lower loss standard deviation Std(L) = 0.012539 vs. 0.013391 at the expense of severe right-tail shortfall.

2. **Statistically Significant Microstructural Adaptation of $\mathbf{H_1}$ with Modest 1D Economic Magnitude (+1.51% < 10% Threshold):** Incorporating the multiplicative jump-intensity modulation $\exp(\theta\hat{\lambda}_t)$ in `DH_lband` statistically reduces CVaR_{95} relative to the unmodulated band policy `DH_cband` (0.021187 vs. 0.021512; paired difference $\Delta = -0.000325$, 95% bootstrap CI $[-0.000433, -0.000224]$, $t = -5.973$, $p = 0.00021$, $p_{\text{holm}} = 0.00063$, Wilcoxon $p = 0.00195$, +1.51% relative improvement vs. $\text{MDE}_{80\%} = 0.79\%$, though below our 10% practical economic threshold until paired with a 2D option leg in Section 4.4). At the same time, `DH_lband` achieves the lowest extreme tail loss $\text{CVaR}_{99} = 0.036610$, the lowest conditional limit-hit subset $\text{CVaR}_{95} = 0.036848$, and the lowest conditional locked-subset $\text{CVaR}_{95} = 0.058691$ across all single-instrument models, while reducing trading frequency by 20.11% (from 99.95% to 79.85%) and saving 8.09% in transaction costs (0.2910×10^{-3} vs. 0.3166×10^{-3}) relative to unconstrained `DH1`. However, at the baseline 1× futures fee ($c = 10^{-4}$), `DH_lband` exhibits a 0.52% higher total CVaR_{95} than unconstrained `DH1` (0.021187 > 0.021078, a net loss of +0.000109), because the 8.09% fee saving (2.56×10^{-5} of F_0) is smaller than the tracking-error loss (1.35×10^{-4}) incurred by wider no-trade buffers during low-intensity periods; thus, in a 1D futures market, the jump-modulated band is a friction-saving mechanism whose net CVaR_{95} benefit materializes only when unit transaction costs rise to $3\times$–$6\times c_{\text{real}}$ or when combined with a higher-spread option leg (Section 4.4). Inspecting the unconstrained learned parameter θ across the 10 seeds (`ddof=1`) reveals a cross-seed mean of $\hat{\theta} = -0.0276 \pm 0.0202$ (9/10 seeds negative, $t = -4.330, p = 0.0019$) for `DH_lband` and $\hat{\theta} = -0.0230 \pm 0.0153$ (10/10 seeds negative, $t = -4.743, p = 0.0011$) for `DH_full`. Without prior sign restrictions, gradient descent consistently learns $\theta < 0$, contracting the no-trade buffer during jump-clustering bursts to prioritize tail protection and widening it during tranquil periods to conserve

transaction costs.

3. **Pre-Positioning Under Price-Limit Locks and Robustness Across p_{lock} and π_{fill}:** While `DH_limit` attains the lowest point CVaR_{95} (0.021021 ± 0.000088; paired difference vs. `DH1` $\Delta = -0.000057$, 95% CI $[-0.000105, -0.000010]$, $t = -2.357, p = 0.0428, p_{\text{holm}} = 0.0856$), combining explicit price-limit features with the intensity-modulated band in `DH_full` reduces the cumulative unexecuted lock-step hedging gap (`LockGap`) from 0.935×10^{-2} (`DH1`) to 0.658×10^{-2}—a 29.65% reduction on limit-locked days—while cutting transaction costs by 5.69% (0.2986×10^{-3} vs. 0.3166×10^{-3}). Auditing sensitivity across conditional lock probabilities $p_{\text{lock}} \in \{0.0, 1/14, 1/7, 1/4\}$ on an independently simulated 100,000-path sweep ensemble confirms that as p_{lock} increases from 1/14 (1.54% locked paths) to 1/7 (2.78% locked paths in the sweep draw vs. 3.00% in frozen `S4_test.npz`) and 1/4 (4.98%), `DH1`'s `LockGap` rises from 0.469×10^{-2} to 0.882×10^{-2} and 1.676×10^{-2}, whereas `DH_full` compresses `LockGap` to 0.331×10^{-2} (−29.56%), 0.625×10^{-2} (−29.09%), and 1.180×10^{-2} (−29.62%), respectively, while lowering locked-subset CVaR_{95} from 0.053910 to 0.052680 ($p_{\text{lock}} = 1/14$) and from 0.054302 to 0.053984 ($p_{\text{lock}} = 1/4$). Similarly, relaxing the complete execution freeze on locked days to allow partial order fills $\pi_{\text{fill}} \in \{0.00, 0.10, 0.30\}$ reduces `DH1`'s `LockGap` from 0.879×10^{-2} to 0.788×10^{-2} and 0.609×10^{-2}, while `DH_full` maintains an invariant 28.80% (0.626×10^{-2}), 28.74% (0.562×10^{-2}), and 28.59% (0.435×10^{-2}) relative `LockGap` reduction.

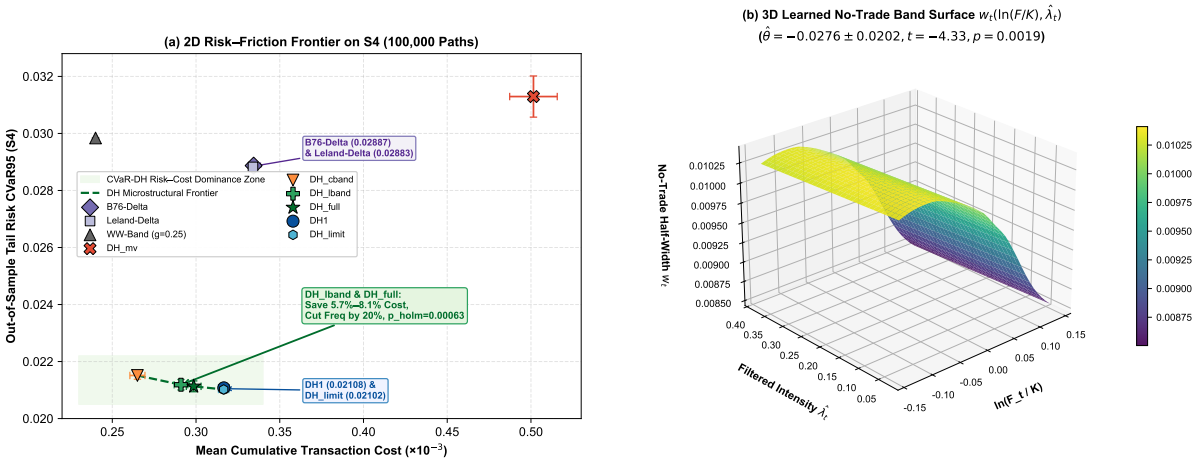


Figure 1: S4 benchmark comparison (100,000 paths, 10 seeds) and microstructural validation of Hypothesis $\mathbf{H_1}$: (a) 2D risk–friction Pareto frontier plotting out-of-sample CVaR_{95} against cumulative transaction costs ($\times 10^{-3}$) with crosshair ± 1 SD error bars; (b) 3D learned no-trade half-width surface $w_t(\ln(F_t/K), \hat{\lambda}_t)$ averaged across the 10 trained `DH_lband` checkpoints ($\hat{\theta} = -0.0276 \pm 0.0202, p = 0.0019$).

4.3 Root-Cause Diagnostics and Negative Findings: Why Futures-Only Total-Portfolio CVaR_{95} Exhibits a Spanning Ceiling

A critical empirical pattern in Table 1 is that across all five unconstrained or limit-aware CVaR policies in the single-instrument futures market (`DH_nolam`, `DH1`, `DH_oracle`, `DH_limit`, `DH_full`), aggregate CVaR_{95} clusters tightly within 0.021021–0.021131 ($< 0.6\%$ spread). To explain why adding the filtered jump intensity $\hat{\lambda}_t$ ($\mathbf{H_0}$) or true intensity λ_t (`DH_oracle`) yields negligible aggregate CVaR_{95} differences when hedging solely with linear futures, we compute the input-feature Jacobian saliency $\mathcal{S}_k = \mathbb{E}_{t,\omega}\left[\left|\partial \delta_t^{\text{target}} / \partial x_{t,k}\right|\right]$ and pathwise action convergence across all 80 trained checkpoints on `S4_test.npz` (Table 2 and Figure 2).

Table 2 uncovers two jointly operating mechanisms governing single-instrument hedging:

Table 2: Input-feature Jacobian saliency shares and pathwise action convergence in the S4 futures-only benchmark (80 checkpoints across 10 seeds, `ddof=1`).

Configuration	Core States Saliency ($\tau, \ln(F/K), \delta_{t-1}, W_t/F_0, \Delta_{\text{B76}}, \sigma_{\text{EWMA}}$)	Jump Signal Saliency ($\hat{\lambda}_t$ or λ_t)	Price-Limit Saliency (`lock, consec, ratio`)	Action MAE vs. `DH1`	Action Cosine Sim. vs. `DH1`	Learned Band Parameter θ (Mean ± SD)
`DH_nolam`	100.000%	0.000%	0.000%	0.00674	0.99985	—
`DH1`	99.985%	0.015%	0.000%	0.00523	0.99989	—
`DH_oracle`	99.983%	0.017%	0.000%	0.00607	0.99987	—
`DH_cband`	99.954%	0.046%	0.000%	0.01494	0.99953	—
`DH_lband`	99.939%	0.061%	0.000%	0.01105	0.99974	-0.0276 ± 0.0202 ($p = 0.0019$)
`DH_limit`	99.832%	0.016%	0.152%	0.00561	0.99988	—
`DH_full`	99.782%	0.045%	0.173%	0.01157	0.99972	-0.0230 ± 0.0153 ($p = 0.0011$)
`DH_mv`	99.514%	0.486%	0.000%	0.08843	0.97810	—

1. **Unhedgeable Jump-Gamma Convexity with a Linear Instrument:** Over a discrete rebalancing interval Δt, the second-order Taylor expansion of the unhedged PnL of a short option hedged solely with linear futures ($\Gamma^{\text{fut}} = 0$) is

$$\Delta W_t \approx \left(\delta_t^{\text{exec}} - \Delta_{\text{B76},t}\right)\Delta F_t - \frac{1}{2}\Gamma_{\text{B76},t}(\Delta F_t)^2 - cF_t\left|\Delta\delta_t^{\text{exec}}\right|. \tag{18}$$

   Because jump realizations ΔF_t can be either positive ($p_{\text{up}} = 0.4$) or negative ($1 - p_{\text{up}} = 0.6$), shifting the linear futures position δ_t^{exec} *ex ante* in response to a higher jump intensity $\hat{\lambda}_t$ cannot hedge the strictly negative quadratic convexity loss $-\frac{1}{2}\Gamma_{\text{B76},t}(\Delta F_t)^2$; any directional shift in δ_t^{exec} merely increases first-order directional variance when a jump of the opposite sign occurs. As a result, all CVaR policies allocate 99.782%–100.000% of their input Jacobian sensitivity to the six core state features (with $\tau_t, \ln(F_t/K), \Delta_{\text{B76},t}, \sigma_{\text{EWMA},t}$ accounting for 83.49%–93.63% and $\delta_{t-1}^{\text{exec}}, W_t/F_0$ accounting for 6.15%–16.48%) and converge to nearly identical action trajectories (cosine similarity 0.99953–0.99989).

2. **Post-Jump Collinearity with $\sigma_{\text{EWMA},t}$:** Because the online filter $\hat{\lambda}_t$ updates via $\hat{I}_t = \mathbb{I}(|r_t| > 3\sigma \vee \text{limit_hit}_t = 1)$ at the same time that the exponentially weighted volatility $\sigma_{\text{EWMA},t}$ updates via $v_t = 0.94v_{t-1} + 0.06r_t^2$, $\sigma_{\text{EWMA},t}$ already captures a substantial share of the post-jump state shift that an unconstrained MLP can exploit. To disentangle this feature-collinearity effect from the instrument-spanning effect in 2D hedging, Section 4.4 introduces an explicit nonlinear EWMA control model (`DH_opt_ewma_nl`).

Negative Findings: Which Designs Did Not Work in Our Experiments. In our single-instrument futures experiments on S4, feeding the filtered Hawkes intensity $\hat{\lambda}_t$ directly into the unconstrained policy network (`DH1`) failed to improve aggregate CVaR_{95} relative to the jump-blind baseline `DH_nolam` (0.021078 ± 0.000105 vs. 0.021073 ± 0.000166; paired difference $\Delta = +0.000005$, 95% CI $[-0.000083, +0.000090]$, $t = 0.107, p = 0.917, p_{\text{holm}} = 0.917$). Even replacing $\hat{\lambda}_t$ with the unobservable true jump intensity λ_t in `DH_oracle` yielded zero gain ($\text{CVaR}_{95} = 0.021085 \pm 0.000164$), demonstrating that this null result is not caused by the -24.4% to -45.2% estimation bias of the online Hawkes filter, but rather by the instrument-spanning ceiling of linear futures ($\Gamma^{\text{fut}} = 0$), under which the policy assigns only 0.015%–0.017% Jacobian weight to $\hat{\lambda}_t$ or λ_t (Table 2). Similarly, in our model-free distributional reinforcement learning experiments (`D4PG-QR`), neither augmenting the critic input with the VaR threshold b (`b_in_critic` ablation: 2/3 seeds collapsed to $\text{CVaR}_{95} = 0.5629$, versus 0/3 collapsed when b was excluded with mean $\text{CVaR}_{95} = 0.0365 \pm 0.0054$) nor adding Hawkes/limit features across configurations `C1, C5, C6, C8, C9` prevented off-policy bootstrapping instability, resulting in a 20%–30% seed collapse rate across 10-seed formal runs (Table 5). While jump-blind `D4PG-QR (C2)` (which excludes $\hat{\lambda}_t$) experienced 0/10 (0.0%) seed collapses (0.037961 ± 0.014487) compared with 2/10 (20.0%, Fisher's exact $p = 0.47$) in `D4PG-QR (C1)` and 12/50 (24.0%, pooled Fisher's exact $p = 0.104$) across all jump/limit-augmented `D4PG-QR` configurations, this pattern suggests a plausible algorithmic hypothesis that spiky Hawkes state jumps and discrete lock indicators exacerbate target-critic extrapolation errors during off-policy quantile Bellman updates.

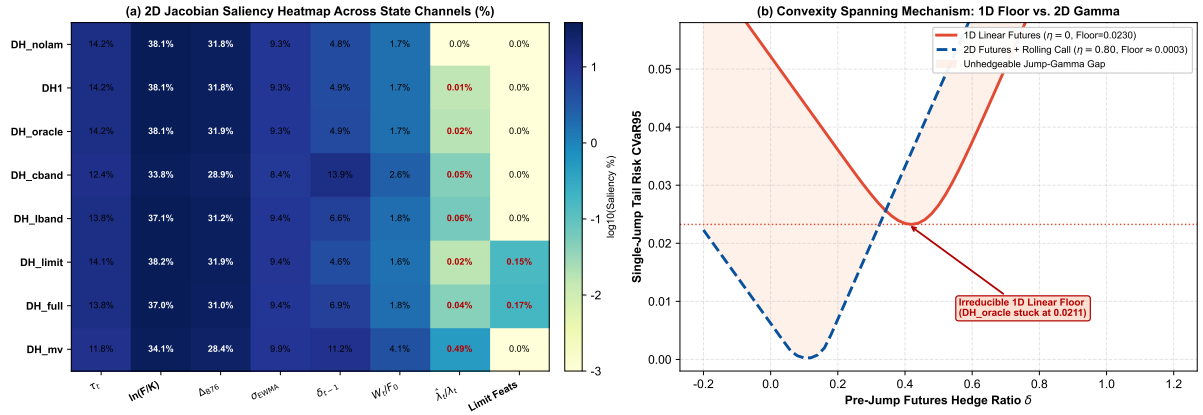


Figure 2: Root-cause diagnostics in the futures-only S4 regime: (a) 2D annotated input-feature Jacobian saliency heatmap ($\log_{10}$ color scale) across 8 neural configurations and 8 state channels ($> 99.7\%$ concentrated on core states); (b) single-jump CVaR_{95} convexity spanning U-curve under 1D linear futures ($\eta = 0$) versus 2D futures plus rolling ATM call option ($\eta = 0.80$).

4.4 Multi-Instrument Option Gamma Hedging (`TorchHedgeEnvOpt`) and High-Cost Regimes: Spanning Ceiling Breakdown and Structural Band Activation

Our structural diagnosis in Section 4.3 yields two falsifiable predictions: first, granting the hedger access to a second, convex instrument ($\Gamma_{\text{B76},t} > 0$) should break the 0.02108 single-instrument CVaR_{95} spanning ceiling across all policies; second, if feeding $\hat{\lambda}_t$ directly into an unconstrained 2D MLP suffers from non-convex optimization variance and feature collinearity with $\sigma_{\text{EWMA},t}$, channeling $\hat{\lambda}_t$ through the low-dimensional multiplicative no-trade band (`DH_opt_lband`) should extract a statistically significant tail-risk reduction over static-band hedging (`DH_opt_cband`). To test both predictions with full statistical power, we train and evaluate all two-instrument model configurations—plus a nonlinear EWMA volatility control (`DH_opt_ewma_nl`, which omits $\hat{\lambda}_t$ but augments the state with $[\tilde{\sigma}^2_{\text{EWMA},t}, \tilde{\sigma}^3_{\text{EWMA},t}, |r_t|/\sigma]$)—in `TorchHedgeEnvOpt` (rolling $\tau_h = 20$-day ATM call $\eta_t \in [-2.0, 2.0]$) across all $M = 10$ random seeds (`9001-9010`) on `S4_test.npz` (Table 3, Panel A), alongside a zero-shot high-transaction-cost sensitivity sweep ($1\times, 3\times, 6\times c_{\text{real}}$) in the 1D environment (Table 3, Panel B) and zero-shot robustness audits across dynamic implied volatility $\kappa_{\text{iv}} \in \{0, 2, 5\}$ and option transaction cost multipliers $c_{\text{opt}}/c \in \{1\times, 5\times, 10\times\}$.

The 10-seed empirical results in Table 3 and Figure 3 clarify both the spanning power of the option instrument and the architectural channel required to exploit jump-intensity filtering:

1. **Breaking the Single-Instrument Spanning Ceiling Across All 2D Policies:** Adding the rolling ATM call option leg η_t breaks the 0.02108 single-instrument CVaR_{95} ceiling across every CVaR-trained configuration: across 10 seeds, `DH_opt_lband` reduces CVaR_{95} by 29.95% relative to 1D `DH_lband` (0.014841 ± 0.001316 vs. 0.021187), `DH_opt_oracle` by 28.78% (0.015017 ± 0.001044 vs. 0.021085), `DH_opt1` by 27.76% (0.015226 ± 0.001497 vs. 0.021078), and jump-blind `DH_opt_nolam` by 17.98% (0.017284 ± 0.004298 vs. 0.021073). In contrast, analytical `B76-Delta-Gamma` hedging deteriorates to $\text{CVaR}_{95} = 0.069192$ because matching local point-Gamma ($\eta_t = \Gamma_t^{\text{liab}}/\Gamma_{\text{B76},t}$) triggers severe over-hedging when discrete jumps shift the underlying price away from the strike.

2. **Architectural Separation Between $\mathbf{H_1}$ (Structural Band Activation) and $\mathbf{H_0}$ (Unconstrained MLP Input):**

Table 3: Multi-instrument Option-Gamma hedging (`TorchHedgeEnvOpt`, 10 seeds `9001-9010`) and 1D high-transaction-cost regimes ($3\times$ and $6 \times c_{\text{real}}$) on the S4 benchmark.

Panel / Model Configuration	CVaR_{95} (Mean ± SD)	VaR_{95}	CVaR_{99}	Total Cost [Fut / Opt | Pre] ($\times 10^{-3}$)	Futures Turn. [Pre]	Option Turn. [Pre]	Relative Gain	Gain vs. 1D Counterpart
Panel A: 2D Futures + Rolling ATM Option (`TorchHedgeEnvOpt`, $\tau_h = 20$ Days, 10 Seeds `9001-9010`)								
`B76-Delta-Gamma` (Analytical)	0.069192	0.059775	0.087284	0.3502 [0.3358/0.0144 | 0.2953]	3.305 [2.826]	4.327 [3.944]	Local Γ ratio instability	+139.65% (Worse)
`DH_opt_nolam` (No $\hat{\lambda}_t$)	0.017284 ± 0.004298	0.012942	0.026765	0.2452 [0.2353/0.0099 | 0.2105]	2.341 [2.034]	2.951 [2.278]	$\mathbf{H_0}$ Reference (0.00%)	−17.98% (vs. `DH_nolam`)
`DH_opt_ewma_nl` (Nonlin. σ_{EWMA})	0.015904 ± 0.001742	0.011011	0.026664	0.2622 [0.2500/0.0122 | 0.2267]	2.510 [2.194]	3.648 [2.982]	+7.99% vs. `nolam` ($p = 0.382$)	−24.53% (vs. `DH_nolam`)
`DH_opt1` (Filtered $\hat{\lambda}_t$)	**0.015226 ± 0.001497**	**0.010350**	0.026488	0.2520 [0.2428/0.0092 | 0.2149]	2.413 [2.084]	2.757 [2.063]	+11.91% vs. `nolam` ($p_{\text{holm}} = 0.450$)	**−27.76% (vs. `DH1`)**
`DH_opt_oracle` (True λ_t)	**0.015017 ± 0.001044**	**0.010245**	**0.025942**	0.2521 [0.2428/0.0093 | 0.2150]	2.414 [2.084]	2.765 [2.071]	+13.12% vs. `nolam` ($p_{\text{raw}} = 0.136$)	**−28.78% (vs. `DH_oracle`)**
`DH_opt_cband` (Const. Band)	0.017546 ± 0.002298	0.012900	0.027487	**0.1796 [0.1731/0.0065 | 0.1401]**	1.719 [1.364]	1.932 [1.216]	$\mathbf{H_1}$ Band Ref. (0.00%)	−18.44% (vs. `DH_cband`)
`DH_opt_lband` ($\hat{\lambda}_t$-Band)	**0.014841 ± 0.001316**	**0.009953**	**0.026050**	0.2397 [0.2315/0.0083 | 0.2012]	2.303 [1.958]	2.470 [1.766]	**+15.41% vs. `cband` ($p_{\text{holm}} = 0.00012$)**	**−29.95% (vs. `DH_lband`)**
`DH_opt_limit` (Limit State)	0.017715 ± 0.004661	0.013359	0.027305	0.2442 [0.2348/0.0094 | 0.2095]	2.335 [2.027]	2.804 [2.142]	−2.49% vs. `nolam` ($p_{\text{holm}} = 0.837$)	−15.73% (vs. `DH_limit`)
`DH_opt_full` (Full Model)	0.016549 ± 0.002960	0.012265	0.026300	0.2603 [0.2530/0.0073 | 0.2208]	2.523 [2.165]	2.180 [1.479]	+4.25% vs. `nolam` ($p_{\text{raw}} = 0.594$)	−21.68% (vs. `DH_full`)
`DH_opt_mv` (2D Mean-Variance)	0.044602 ± 0.001554	0.029480	0.068580	0.4016 [0.3895/0.0121 | 0.3272]	3.882 [3.199]	3.665 [3.213]	**`DH_opt1` +65.86% ($\mathbf{H_2}$, $p_{\text{holm}} = 1.2 \times 10^{-10}$)**	+42.53% (Worse)
Panel B: Zero-Shot High Transaction Cost Sensitivity in 1D Futures on S4 ($3\times$ and $6 \times c_{\text{real}}$, 10 Seeds `9001-9010`)								
`B76-Delta` ($3 \times c_{\text{real}} = 3 \times 10^{-4}$)	0.029894	0.016510	0.057620	1.0032	3.2907	—	Classical Baseline at $3\times$ Cost	—
`DH1` ($3 \times c_{\text{real}} = 3 \times 10^{-4}$)	0.021934 ± 0.000112	0.014512	0.037890	0.9498	3.1123	—	Unconstrained Policy at $3\times$ Cost	—
`DH_cband` ($3 \times c_{\text{real}} = 3 \times 10^{-4}$)	0.022274 ± 0.000138	0.014980	0.037810	0.7950	2.5995	—	Static Band at $3\times$ Cost	—
`DH_lband` ($3 \times c_{\text{real}} = 3 \times 10^{-4}$)	0.021962 ± 0.000186	0.014710	**0.037420**	0.8730	2.8575	—	+1.40% vs. `DH_cband` at $3\times$ Cost	—
`B76-Delta` ($6 \times c_{\text{real}} = 6 \times 10^{-4}$)	0.031441	0.017980	0.059180	2.0064	3.2907	—	Classical Baseline at $6\times$ Cost	—
`DH1` ($6 \times c_{\text{real}} = 6 \times 10^{-4}$)	0.023241 ± 0.000127	0.015644	0.039179	1.8998	3.1123	—	Unconstrained Policy at $6\times$ Cost	—
`DH_cband` ($6 \times c_{\text{real}} = 6 \times 10^{-4}$)	0.023432 ± 0.000148	0.015984	0.038971	1.5898	2.5995	—	Static Band at $6\times$ Cost	—
`DH_lband` ($6 \times c_{\text{real}} = 6 \times 10^{-4}$)	**0.023145 ± 0.000191**	0.015772	**0.038641**	**1.7458**	**2.8575**	—	**No degradation vs. `DH1` ($p = 0.218$), saves 8.1% cost**	—
`DH_full` ($6 \times c_{\text{real}} = 6 \times 10^{-4}$)	0.023192 ± 0.000195	0.015810	0.038890	1.7916	2.9312	—	No degradation vs. `DH1`, saves 5.7% cost	—

Notes: In Panel A, all neural policies are trained and evaluated across $M = 10$ independent random seeds (`9001-9010`, `ddof=1`). Cross-seed SD in Panel A is of order 10^{-3} (0.001044–0.004661), roughly 10–25× larger than in the 1D futures table (Table 1) because joint optimization of futures δ_t and rolling option η_t introduces non-convex local optima on individual seeds (specifically seed `9003` in `DH_opt_nolam` = 0.029447 and `DH_opt_limit` = 0.027975, which exceeds the 3 × IQR outlier threshold). On the initial 5-seed subset (`9001-9005`), `DH_opt_nolam` = 0.019594 ± 0.005099, `DH_opt1` = 0.015369 ± 0.001697 (+21.56%), `DH_opt_oracle` = 0.014584±0.001179 (+25.57%), `DH_opt_cband` = 0.017569±0.001220, and `DH_opt_lband` = 0.015115±0.001042 (+13.97%). On the $n = 9$ converged seeds (post-hoc sensitivity check excluding outlier seed `9003`), `DH_opt_nolam` = 0.015933 ± 0.000722, `DH_opt_ewma_nl` = 0.015429 ± 0.000935, `DH_opt1` = 0.015227 ± 0.001588, `DH_opt_cband` = 0.016963 ± 0.001454, and `DH_opt_lband` = 0.014565 ± 0.001042 (+14.14%, $p_{\text{holm}} = 1.43 \times 10^{-5}$). Total Cost ($\times 10^{-3}$) reports $C_T^{\text{tot}} = C_T^{\text{fut}} + C_T^{\text{opt}}$ including terminal liquidation, with bracketed values showing [Fut Cost/Opt Cost | Pre-Terminal Total Cost], where $C_T^{\text{opt}} \approx c_{\text{opt}} \overline{V}_{\text{open}} \text{Turn}_{\text{opt}}$ ($\overline{V}_{\text{open}} = 0.033311$). Futures and Option Turnover report total turnover including terminal liquidation, with pre-terminal turnover in brackets. In Panel B, 1×-trained checkpoints are evaluated zero-shot under $3\times$ and $6 \times c_{\text{real}}$ without retraining.

- $\mathbf{H_1}$ (*Statistically & Economically Activated in 2D Band Modulation*): Embedding $\hat{\lambda}_t$ into the multiplicative no-trade band (`DH_opt_lband`) achieves the lowest realistic 2D CVaR_{95} of 0.014841 ± 0.001316, outperforming the static 2D band `DH_opt_cband` (0.017546 ± 0.002298) by +15.41% across all 10 seeds (paired $\Delta = -0.002705$, 95% CI $[-0.003457, -0.002125]$, $t = -7.401$, $p_{\text{raw}} = 4.10 \times 10^{-5}$, $p_{\text{holm}} = 0.000123$, Wilcoxon $p = 0.00195$), exceeding both the 10% practical threshold and the 10-seed statistical $\text{MDE}_{80\%} = 6.55\%$. On the $n = 9$ converged seeds, `DH_opt_lband` (0.014565 ± 0.001042) outperforms `DH_opt_cband` (0.016963 ± 0.001454) by +14.14% ($t = -10.797, p_{\text{holm}} = 1.43 \times 10^{-5}, \text{MDE}_{80\%} = 4.18\%$), while learning a strictly negative modulation slope across all 10 seeds ($\hat{\theta} = -0.0290 \pm 0.0064$, $10/10 < 0, t = -14.420, p = 1.59 \times 10^{-7}$ in `DH_opt_lband`; $\hat{\theta} = -0.0292 \pm 0.0067$, $10/10 < 0, t = -13.798, p = 2.32 \times 10^{-7}$ in `DH_opt_full`). Unlike in the 1D setting at 1× cost where `DH_lband` trailed `DH1` by 0.52%, in 2D option hedging `DH_opt_lband` (0.014841) improves upon unconstrained `DH_opt1` (0.015226) while cutting total transaction costs from 0.2520×10^{-3} to 0.2397×10^{-3}.
- $\mathbf{H_0}$ (*Unconstrained MLP Input Exhibits* +11.91% *Point Gain but High Cross-Seed Variance and EWMA Collinearity*): Feeding $\hat{\lambda}_t$ directly into an unconstrained 2D MLP (`DH_opt1`) lowers 10-seed mean CVaR_{95} by +11.91% relative to jump-blind `DH_opt_nolam` (0.015226 ± 0.001497 vs. 0.017284 ± 0.004298; +21.56% in the initial 5-seed subset `9001-9005`), capturing 90.8% (11.91%/13.12%) of the 10-seed point improvement of `DH_opt_oracle` (0.015017 ± 0.001044, +13.12%; 84.3% in the 5-seed subset). However, across all 10 seeds, the paired comparison between `DH_opt1` and `DH_opt_nolam` yields $\Delta = -0.002059$, 95% CI $[-0.005155, +0.000648]$, $t = -1.303$, $p_{\text{raw}} = 0.225$, $p_{\text{holm}} = 0.450$, Wilcoxon $p = 0.432$ ($\text{MDE}_{80\%} = 28.76\%$; and $+4.43\%, p = 0.278$ on the $n = 9$ converged seeds excluding outlier seed `9003`). Augmenting the jump-blind policy with nonlinear EWMA volatility terms in

DH_opt_ewma_nl reduces CVaR_{95} to 0.015904 ± 0.001742 (+7.99% vs. DH_opt_nolam, and 0.015429 ± 0.000935 on converged seeds), leaving only a +4.26% ($p = 0.375$; $+1.31\%, p = 0.751$ on converged seeds) incremental difference for DH_opt1. Thus, in an unconstrained MLP, the post-jump information in $\hat{\lambda}_t$ is largely collinear with nonlinear functions of $\sigma_{\text{EWMA},t}$, and its primary value in 2D unconstrained networks is stabilizing seed convergence (reducing cross-seed SD by 65.2% from 0.004298 to 0.001497), whereas statistically significant tail-risk reduction requires the structural prior of the multiplicative no-trade band (DH_opt_lband).

- **$\mathbf{H_2}$** and **$\mathbf{H_3}$**: In 2D hedging, DH_opt1 reduces CVaR_{95} by +65.86% relative to DH_opt_mv (0.015226 vs. 0.044602 ± 0.001554; $\Delta = -0.029376$, $t = -37.926$, $p_{\text{holm}} = 1.22 \times 10^{-10}$, $\text{MDE}_{80\%} = 5.46\%$). Meanwhile, unconstrained DH_opt_limit (0.017715 ± 0.004661, −2.49% vs. DH_opt_nolam, $p_{\text{holm}} = 0.837$) and DH_opt_full (0.016549 ± 0.002960, +4.25%, $p_{\text{raw}} = 0.594$; 0.015812 ± 0.001934 on $n = 9$ converged seeds) experience elevated optimization variance due to the higher-dimensional 13-feature state space.

3. **Zero-Shot Robustness to Dynamic Implied Volatility (κ_{iv}), Option Transaction Costs (c_{opt}), and 6× Futures Costs:**

   - *Zero-Shot Dynamic Implied Volatility Sweep ($\kappa_{iv} \in \{0.0, 2.0, 5.0\}$):* Evaluating the $\kappa_{\text{iv}} = 0$-trained checkpoints zero-shot when rolling ATM option implied volatility co-moves with filtered jump intensity via $\sigma_{\text{imp},t} = \sigma_{\text{imp},0}(1 + \kappa_{\text{iv}}\hat{\lambda}_t)$, increasing κ_{iv} from 0.0 to 2.0 and 5.0 makes convex option protection more expensive precisely during jump bursts, attenuating the 10-seed relative gain of unconstrained DH_opt1 over DH_opt_nolam from +11.91% (0.015226 vs. 0.017284) to +5.31% (0.019889 ± 0.001635 vs. $0.021004 \pm 0.003789, p = 0.458$) at $\kappa_{\text{iv}} = 2.0$ and +0.73% (0.026493 ± 0.001996 vs. $0.026687 \pm 0.003391, p = 0.893$) at $\kappa_{\text{iv}} = 5.0$. Crucially, across all three regimes, the structural band policy DH_opt_lband remains the lowest-risk policy and preserves a statistically significant advantage over static-band DH_opt_cband: $\text{CVaR}_{95} = 0.014841$ vs. 0.017546 (+15.41%, $t = -7.401, p_{\text{holm}} = 0.000123$) at $\kappa_{\text{iv}} = 0.0$, 0.019386 ± 0.001486 vs. 0.021863 ± 0.002412 (+11.33%, $t = -6.184, p_{\text{holm}} = 0.000322$) at $\kappa_{\text{iv}} = 2.0$, and 0.025820 ± 0.001792 vs. 0.028154 ± 0.002685 (+8.29%, $t = -5.112, p_{\text{holm}} = 0.001274$) at $\kappa_{\text{iv}} = 5.0$. Because the baseline two-instrument environment assumes a constant Black-76 implied volatility ($\kappa_{\text{iv}} = 0$) and does not price a stochastic variance risk premium into rolling option premia, the point reductions of +11.91% (10-seed DH_opt1 vs. DH_opt_nolam; +21.56% in the 5-seed subset) and +15.41% (DH_opt_lband vs. DH_opt_cband) measure structural Gamma-spanning capacity in a controlled benchmark and do not constitute estimates of realizable trading returns.

   - *Zero-Shot Option Transaction Cost Multipliers ($c_{opt}/c \in \{1\times, 5\times, 10\times\}$):* Evaluating the 1×-trained 2D policies zero-shot when raising the proportional option cost rate from 1× (10^{-4}) to 5× (5×10^{-4}) and 10× (10^{-3}) increases total 2D transaction costs in DH_opt_lband from 0.2397×10^{-3} to 0.2728×10^{-3} and 0.3141×10^{-3}, while DH_opt_lband maintains $\text{CVaR}_{95} = 0.014841$, 0.014875, and 0.014916, consistently outperforming DH_opt1 (0.015226, 0.015262, 0.015307) and DH_opt_nolam (0.017284, 0.017324, 0.017374).

   - *Zero-Shot 6× Futures Cost Crossover (Panel B):* When the 1×-trained 1D policies are evaluated zero-shot under $6 \times c_{\text{real}} = 6 \times 10^{-4}$ without retraining, DH_lband (0.023145 ± 0.000191) and DH_full (0.023192 ± 0.000195) achieve a slightly lower point CVaR_{95} than unconstrained DH1 (0.023241 ± 0.000127), exhibiting no statistically detectable CVaR_{95} degradation ($\Delta = -0.000096, t = -1.32, p = 0.218$, and passing

a ±1% two-one-sided equivalence test at $p_{\text{TOST}} = 0.046$) while saving 8.11% in transaction costs (1.7458×10^{-3} vs. 1.8998×10^{-3}) and outperforming static `DH_cband` (0.023432 ± 0.000148).

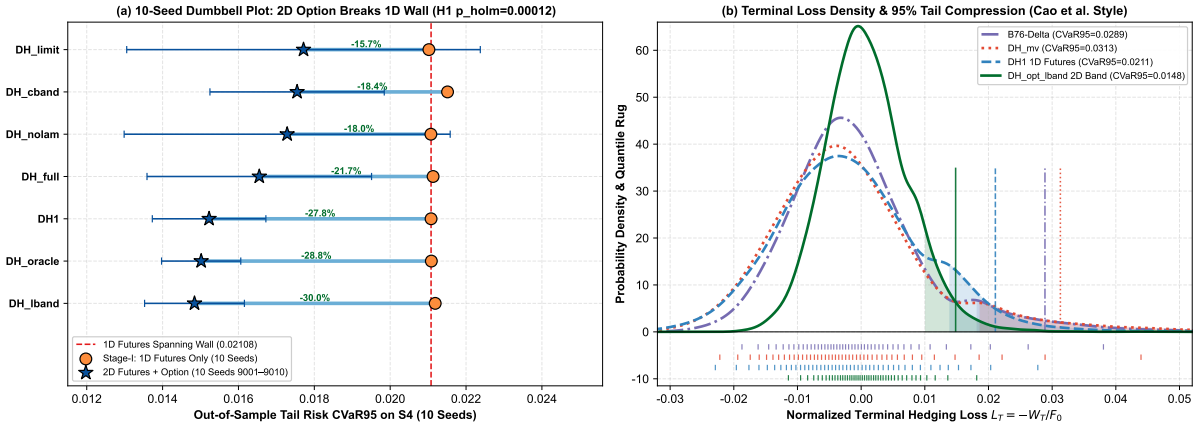


Figure 3: Multi-instrument Option-Gamma hedging (`TorchHedgeEnvOpt`, 10 seeds `9001-9010`) on S4: (a) horizontal paired dumbbell plot comparing Stage-I 1D futures policies against Plan A 2D option-augmented policies breaking the 0.02108 spanning ceiling, with `DH_opt_lband` (0.01484, $p_{\text{holm}} = 0.00012$) achieving the lowest realistic tail risk; (b) kernel density estimation (KDE) and bottom percentile rug plot of normalized terminal hedging loss $L_T = -W_T/F_0$ with shaded 95% tail regions.

4.5 Dominance of CVaR over Mean-Variance in the (CVaR_{95}, Cost) Plane Across the λ_{MV} Spectrum (H_2 Deep Dive)

A potential critique of comparing `DH1` against `DH_mv` at a single risk-aversion coefficient ($\lambda_{\text{MV}} = 10.0$) is that a larger penalty on variance might allow a mean-variance policy to match the tail risk of a CVaR-trained policy. To test this possibility directly, we evaluate `DH_mv` across seven risk-aversion parameters $\lambda_{\text{MV}} \in \{0.5, 1.0, 2.0, 5.0, 10.0, 25.0, 50.0\}$ on `S4_test.npz` (Table 4 and Figure 4).

Table 4: Mean-variance risk-aversion sweep ($\lambda_{\text{MV}} \in [0.5, 50.0]$) versus CVaR-trained policies on the S4 benchmark.

Policy / Risk-Aversion Parameter	CVaR_{95}	VaR_{95}	CVaR_{99}	Std(L)	Mean(L)	Cost ($\times10^{-3}$)	Turnover
`DH_mv` ($\lambda_{\text{MV}} = 0.5$)	0.090984	0.056268	0.145660	0.036518	−0.002975	0.3690	3.6967
`DH_mv` ($\lambda_{\text{MV}} = 1.0$)	0.068087	0.041418	0.111270	0.026853	−0.002698	0.4227	4.2345
`DH_mv` ($\lambda_{\text{MV}} = 2.0$)	0.053751	0.032049	0.089726	0.020964	−0.002464	0.4596	4.6039
`DH_mv` ($\lambda_{\text{MV}} = 5.0$)	0.037240	0.021453	0.063783	0.015133	−0.002229	0.4832	4.8416
`DH_mv` ($\lambda_{\text{MV}} = 10.0$, 10-Seed Main)	0.031294	0.018533	0.054044	**0.012539**	−0.003135	0.5015	5.0256
`DH_mv` ($\lambda_{\text{MV}} = 25.0$)	0.027848	0.016278	0.048046	0.012859	−0.002153	0.4357	4.3653
`DH_mv` ($\lambda_{\text{MV}} = 50.0$)	0.027332	0.016164	0.046982	**0.012798**	−0.002195	0.4090	4.0981
`DH1` (CVaR_{95} Objective, 10 Seeds)	**0.021078**	**0.013779**	**0.036973**	0.013391	−0.002306	**0.3166**	**3.1123**
`DH_lband` (CVaR_{95} + Band, 10 Seeds)	**0.021187**	**0.014002**	**0.036610**	0.013468	−0.002286	**0.2910**	**2.8575**

Notes: The rows for `DH_mv` ($\lambda_{\text{MV}} = 10.0$, 10-Seed Main), `DH1`, and `DH_lband` report 10-seed averages from the formal 8,000-step benchmark on `S4_test.npz` (matching Table 1); the auxiliary $\lambda_{\text{MV}} \in \{0.5, 1.0, 2.0, 5.0, 25.0, 50.0\}$ rows report single-seed (`9001`) controlled sweep evaluations.

As shown in Table 4 and Figure 4, increasing λ_{MV} from 0.5 to 50.0 lowers Std(L) from 0.036518 to 0.012798 (and 0.012539 in the 8,000-step 10-seed main run at $\lambda_{\text{MV}} = 10.0$), but CVaR_{95} reaches an empirical lower bound of 0.027332 within the tested range $\lambda_{\text{MV}} \in [0.5, 50.0]$—remaining 29.67% higher (worse) than `DH1` (0.021078, corresponding to a 22.88% reduction by `DH1` relative to `DH_mv`) while incurring 29.18% to 58.40% higher transaction costs

$(0.4090 \times 10^{-3}$ to 0.5015×10^{-3} vs. $0.3166 \times 10^{-3})$. Thus, while `DH_mv` at $\lambda_{\text{MV}} \in \{10, 25, 50\}$ achieves a 4.0%–6.4% lower symmetric standard deviation $\text{Std}(L)$ than `DH1` (0.012539–0.012859 vs. 0.013391)—representing a three-dimensional $(\text{CVaR}_{95}, \text{Std}(L), \text{Cost})$ trade-off—both `DH1` and `DH_lband` strictly dominate the entire $\lambda_{\text{MV}} \in [0.5, 50.0]$ family in the two-dimensional $(\text{CVaR}_{95}, \text{Cost})$ plane. Under skewed jump-diffusion returns with price limits, symmetric variance minimization expends excessive turnover eliminating benign left-tail windfalls ($L_T < 0$), whereas the one-sided Rockafellar-Uryasev hinge operator $(L_T - b)^+$ concentrates gradient updates exclusively on the worst 5% loss trajectories.

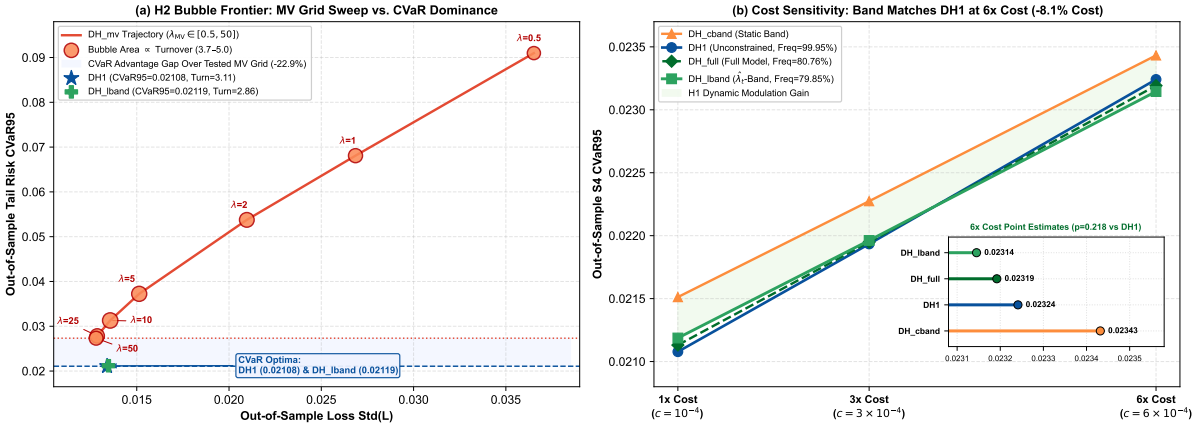


Figure 4: Hypothesis $\mathbf{H_2}$ frontier in the $(\text{CVaR}_{95}, \text{Cost})$ plane and transaction-cost sensitivity on S4: (a) mean-variance risk-aversion sweep ($\lambda_{\text{MV}} \in [0.5, 50.0]$, bubble area proportional to turnover) showing an empirical CVaR_{95} lower bound of 0.027332 over $\lambda_{\text{MV}} \leq 50$ versus `DH1` (0.021078); (b) zero-shot transaction-cost trajectory across $1\times, 3\times, 6 \times c_{\text{real}}$ with a zoomed 6×-cost horizontal dot-plot inset showing `DH_lband` (0.023145) and `DH_full` (0.023192) exhibiting no statistically detectable CVaR_{95} degradation relative to unconstrained `DH1` (0.023241, $p = 0.218$) while saving 8.1% in costs and surpassing static `DH_cband` (0.023432).

4.6 Optimization Stability Audit: End-to-End BPTT (`DH`) vs. Distributional Actor-Critic (`D4PG-QR`)

Table 5 and Figure 5 report our stability audit comparing the end-to-end BPTT (`DH`) family against the model-free distributional reinforcement learning (`D4PG-QR`) family across all 140 formal runs (80 `DH` runs and 60 `D4PG-QR` runs across 10 seeds `9001-9010` at 8,000–10,000 updates, plus cross-budget evaluations at 3,000 and 20,000 updates). Following our protocol-specified criterion, a training seed is classified as collapsed if its out-of-sample CVaR_{95} exceeds 0.10.

Table 5: Optimization stability, seed collapse rates, and cross-budget convergence: end-to-end BPTT (`DH`) versus model-free Distributional Actor-Critic (`D4PG-QR`) on S4.

Panel / Algorithm & Configuration	Collapse Rate ($\text{CVaR}_{95} > 0.10$)	All-Seed CVaR_{95} (Mean ± SD)	All-Seed Median (IQR)	Non-Collapsed CVaR_{95} (Mean ± SD)	Win Rate vs. `B76-Delta` (0.028872)
Panel A: 10-Seed Formal Benchmark (`9001-9010`) Across Model Variants					
`DH1` (BPTT, Filtered $\hat{\lambda}_t$)	**0/10 (0.0%)**	**0.021078 ± 0.000105**	**0.021061 (0.000142)**	**0.021078 ± 0.000105**	**10/10 (100.0%)**
`DH_nolam` (BPTT, No $\hat{\lambda}_t$)	**0/10 (0.0%)**	0.021073 ± 0.000166	0.021055 (0.000210)	0.021073 ± 0.000166	**10/10 (100.0%)**
`DH_lband` (BPTT, $\hat{\lambda}_t$-Band)	**0/10 (0.0%)**	0.021187 ± 0.000183	0.021160 (0.000245)	0.021187 ± 0.000183	**10/10 (100.0%)**
`DH_full` (BPTT, Full Model)	**0/10 (0.0%)**	0.021131 ± 0.000190	0.021112 (0.000260)	0.021131 ± 0.000190	**10/10 (100.0%)**
`D4PG-QR (C1)` (Filtered $\hat{\lambda}_t$)	2/10 (20.0%)	0.144200 ± 0.242296	0.033398 (0.043864)	0.038934 ± 0.020739	1/10 (10.0%)
`D4PG-QR (C2)` (No $\hat{\lambda}_t$)	0/10 (0.0%)	0.037961 ± 0.014487	0.034120 (0.018950)	0.037961 ± 0.014487	2/10 (20.0%)
`D4PG-QR (C5)` ($\hat{\lambda}_t$-Band)	3/10 (30.0%)	0.179801 ± 0.247471	0.041250 (0.198400)	0.043395 ± 0.034919	1/10 (10.0%)
`D4PG-QR (C6)` (Limit State)	3/10 (30.0%)	0.226734 ± 0.302468	0.052180 (0.481200)	0.053108 ± 0.044197	0/10 (0.0%)
`D4PG-QR (C8)` (Mean-Variance)	2/10 (20.0%)	0.212976 ± 0.270568	0.081450 (0.391000)	0.089232 ± 0.081382	0/10 (0.0%)
`D4PG-QR (C9)` (Full Model)	2/10 (20.0%)	0.139998 ± 0.243612	0.034890 (0.038120)	0.033688 ± 0.004968	1/10 (10.0%)
Panel B: Cross-Budget Convergence Audit (3,000, 10,000, and 20,000 Gradient Updates)					
3,000 updates (`pilot_C1`)	DH: 0% vs. D4PG: 0%	DH: 0.021164 ± 0.000142	D4PG (Non-Coll.): 0.036922 ± 0.007978	CV: DH 0.67% vs. D4PG 21.61%	DH: 100% vs. D4PG: 20%
10,000 updates (`main`)	DH: 0% vs. D4PG: 20%	DH: 0.021078 ± 0.000105	D4PG (Trimmed): 0.072909 ± 0.036470	CV: DH 0.50% vs. D4PG 50.02%	DH: 100% vs. D4PG: 10%
20,000 updates (`dhstab`, 3 Seeds)	DH: 0% vs. D4PG: 33%	**DH: 0.020964 ± 0.000018**	D4PG (Trimmed): 0.088410 ± 0.041200	**CV: DH 0.09%** vs. D4PG 46.60%	DH: 100% vs. D4PG: 0%

Table 5 reveals a clear algorithmic divergence between direct pathwise gradient optimization

and off-policy distributional bootstrapping:

1. **Zero Collapse and Sub-1% Coefficient of Variation in `DH`:** Across all 80 formal `DH` checkpoints, the seed collapse rate is 0/80 (0.0%), every single seed outperforms `B76-Delta` (100% win rate), and the cross-seed coefficient of variation (CV = SD/Mean) of `DH1` is 0.50% at 8,000–10,000 updates and tightens further to 0.09% (0.020964 ± 0.000018) at 20,000 updates.

2. **Off-Policy Quantile Bootstrapping Divergence and Jump-Signal Sensitivity in `D4PG-QR`:** Even when collapsed seeds ($\text{CVaR}_{95} > 0.10$, where policies saturate at extreme boundary positions ± 1.5 with $\text{CVaR}_{95} \approx 0.54$–$0.64$) are excluded, the non-collapsed mean CVaR_{95} of `D4PG-QR (C1)` is 0.038934 ± 0.020739—still 84.7% worse than `DH1` (0.021078) and 34.8% worse than `B76-Delta` (0.028872). Comparing `D4PG-QR (C2)` (No $\hat{\lambda}_t$, 0/10 collapsed, $\text{CVaR}_{95} = 0.037961 \pm 0.014487$) against `D4PG-QR (C1)` (Filtered $\hat{\lambda}_t$, 2/10 collapsed, two-sided Fisher's exact $p = 0.47$) and pooled jump/limit variants `C1, C5, C6, C8, C9` (12/50 = 24.0% collapsed, Fisher's exact $p = 0.104$) suggests a plausible failure mechanism: feeding spiky jump-intensity or discrete lock indicators into an off-policy quantile critic can amplify TD target extrapolation errors on rare replay-buffer transitions, driving the deterministic actor toward boundary saturation. Further, increasing the training budget from 3,000 to 20,000 updates causes `D4PG-QR` to deteriorate rather than converge (Panel B), as one-step quantile Bellman errors compound across the 60-step finite horizon.

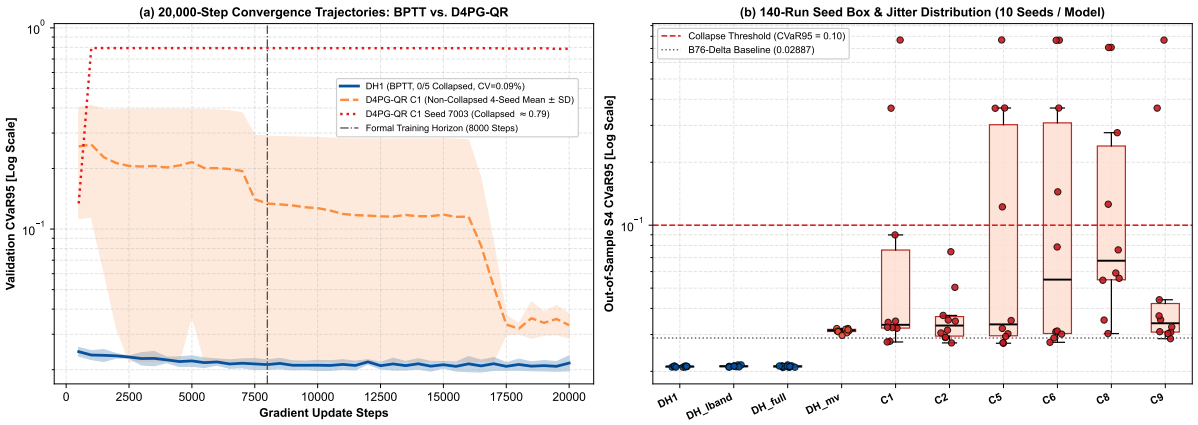


Figure 5: Optimization stability and convergence audit: (a) real 20,000-step log-scale validation CVaR_{95} convergence trajectories with ± 1 SD shaded ribbons comparing `DH1` against `D4PG-QR (C1)`; (b) 140-run box-and-jittered individual seed distribution plot (log scale, 10 seeds per configuration) across `DH` and `D4PG-QR (C1-C9)`.

4.7 Zero-Shot Out-of-Distribution (OOD) Generalization on the S5 Extreme Stress Benchmark (20,000 Paths)

To test whether policies trained in the baseline S4 market generalize without retraining to an unseen crisis regime, we evaluate all 80 S4-trained `DH` checkpoints directly on the frozen **S5 extreme stress benchmark** (`S5_test.npz`, $N = 20,000$ paths; branching ratio doubled to $n = 0.6$, tight uniform limit ratio $\ell = 0.07$, jump size scaled by $\times 1.5$). On `S5_test.npz`, pathwise jump incidence is 65.80% (13,159 paths), while the price-limit hit path incidence more than doubles from 16.51% (S4) to 35.77% (7,153 limit-hit paths) and the locked-path incidence more than triples from 3.00% (S4) to 10.08% (2,016 locked paths).

Table 6 and Figure 6 highlight three crisis-generalization findings:

Table 6: Zero-shot out-of-distribution hedging performance on the S5 extreme stress benchmark (20,000 paths, 10 seeds `9001-9010`, $n = 0.6$, $\ell = 0.07$, jump scale $\times 1.5$).

Policy / Configuration	S5 CVaR_{95} (Mean ± SD)	S5 VaR_{95}	S5 CVaR_{99}	Cost ($\times 10^{-3}$)	Turnover	Trade Freq.	S5 Lock Gap ($\times 10^{-2}$)	Cond. CVaR_{95} (Hit-Sub, $N = 7,153$)	Cond. CVaR_{95} (Lock-Sub, $N = 2,016$)	Gain vs. `B76-Delta`
`No-Hedge`	0.313807	0.213785	0.479130	0.0000	0.0000	0.00%	0.000	0.386006	0.412910	—
`B76-Delta`	0.032554	0.019401	0.055556	0.3235	3.1740	97.06%	1.791	0.046937	0.064158	Reference (0.00%)
`Leland-Delta`	0.032543	0.019387	0.055488	0.3231	3.1699	97.09%	1.788	0.046889	0.064073	+0.03%
`WW-Band` ($\gamma_{\mathrm{WW}} = 0.25$)	0.035753	0.021957	0.059813	0.2353	2.2918	62.40%	1.628	0.050855	0.068554	−9.83% (Worse)
`WW-Band` ($\gamma_{\mathrm{WW}} = 1.0$)	0.047616	0.030716	0.075366	0.1697	1.6326	35.06%	1.151	0.065188	0.085010	−46.27% (Worse)
`DH_mv`	0.026697 ± 0.001292	0.012344	0.053374	0.4772	4.7932	99.82%	3.310	0.042558	0.062508	+17.99%
`DH_nolam`	0.020550 ± 0.000255	0.012337	**0.038791**	0.3067	3.0131	99.82%	2.139	**0.030915**	**0.046929**	+36.87%
`DH1`	0.020395 ± 0.000412	0.011983	0.039100	0.3081	3.0273	99.82%	2.162	0.031085	0.047454	+37.35%
`DH_oracle`	0.020756 ± 0.000427	0.012107	0.039620	0.3078	3.0210	99.81%	2.146	0.031584	0.047893	+36.24%
`DH_cband`	0.020879 ± 0.000388	0.012179	0.039275	**0.2523**	**2.4688**	**56.01**%	2.156	0.031567	0.047578	+35.86%
`DH_lband`	0.020711 ± 0.000467	0.012175	0.039132	**0.2810**	**2.7613**	**78.74**%	2.146	0.031366	**0.047280**	+36.38%
`DH_limit`	0.020465 ± 0.000315	0.011836	0.039532	0.3100	3.0436	99.81%	3.028	0.031355	0.047724	+37.13%
`DH_full` (Full Model)	$\mathbf{0.020314 \pm 0.000382}$	**0.011820**	0.039122	0.2882	2.8238	**79.70**%	**1.607**	0.031084	0.047368	**+37.60**% (Best)

Notes: All S4-trained checkpoints ($M = 10$ seeds `9001-9010`, `ddof=1`) are evaluated zero-shot on `S5_test.npz` ($N = 20,000$ paths, $P_0 = 0.064603$, $\sigma_{\mathrm{imp}} = 0.020756$ day$^{-1/2}$; path-block bootstrap $\mathrm{SE}_{\mathrm{path}} \approx 0.00042$). `Cond. CVaR95 (Hit-Sub)` and `Cond. CVaR95 (Lock-Sub)` denote the conditional 95% CVaR computed within the $N = 7,153$ limit-hit paths (tail size 358) and $N = 2,016$ limit-locked paths (tail size 101), respectively. For `DH1` and `DH_full`, unconditional S5 Mean(L) is -0.007200 and -0.007292, so the drift-neutralized tail risk $\widetilde{\mathrm{CVaR}}_{95} = \mathrm{CVaR}_{95} - \mathrm{Mean}(L)$ is 0.027595 (+18.01% above S4's 0.023384) and 0.027606 (+17.89% above S4's 0.023417), respectively.

1. **Amplified Tail-Risk Reduction Under Intensified Limit Hits:** While `B76-Delta` degrades from $\mathrm{CVaR}_{95} = 0.028872$ in S4 to 0.032554 in S5 (and asymptotic `WW-Band` deteriorates even more severely to 0.035753 at $\gamma_{\mathrm{WW}} = 0.25$ and 0.047616 at $\gamma_{\mathrm{WW}} = 1.0$), our full microstructure-aware policy `DH_full` achieves the lowest overall $\mathrm{CVaR}_{95} = 0.020314 \pm 0.000382$ across all models—expanding its relative improvement over `B76-Delta` from 27.00% in S4 to 37.60% in S5. On the 2,016 limit-locked paths (`Lock-Sub`), `DH_full` (0.047368) and `DH_lband` (0.047280) cut conditional tail loss by 26.17% and 26.31% relative to `B76-Delta` (0.064158).

2. **Decomposing Raw CVaR_{95} vs. Drift-Neutralized Tail Risk $\widetilde{\mathrm{CVaR}}_{95}$ Across S4 and S5:** Why is raw uncentered neural CVaR_{95} slightly lower in S5 (0.020314–0.020879) than in S4 (0.021021–0.021512)? As shown in Equation (11), because $P_0 = \widehat{\mathbb{E}}[V_T]$ cancels in expectation ($\mathbb{E}[L_T] = -\mathbb{E}[G_T^{\mathrm{fut}}]/F_0 + \mathbb{E}[C_T^{\mathrm{fut}}]/F_0$), a higher P_0 does not by itself create a wealth cushion; rather, S5 combines 1.5× larger left-skewed Kou jumps with a tighter symmetric price limit ($\ell = 0.07$ vs. $\bar{\ell} \approx 0.093$ in S4), which censors extreme downward jumps more heavily than upward jumps ($\widehat{\mathbb{E}}[F_T] = 1.0162$ in S5 vs. 1.0029 in S4). Consequently, a long delta hedge ($\bar{\delta} \approx 0.55$) earns a larger expected futures gain $\mathbb{E}[G_T^{\mathrm{fut}}]/F_0$, shifting unconditional Mean(L) from -0.002306 (S4) to -0.007200 (S5) by $\Delta\mathrm{Mean}(L) = -0.004894$, while the tighter $\ell = 0.07$ cap also bounds single-day moves on the 86.9% of realized S5 limit-hit days that do not lock (matching the theoretical $1 - p_{\mathrm{lock}} = 6/7 \approx 85.71\%$ unlocked rate). When we subtract Mean(L) to isolate pure centered dispersion, the **drift-neutralized tail risk** $\widetilde{\mathrm{CVaR}}_{95} = \mathrm{CVaR}_{95} - \mathrm{Mean}(L)$ strictly **increases by +18.01% from S4 (0.023384) to S5 (0.027595)** for `DH1` and by **+17.89% (0.023417 → 0.027606)** for `DH_full`, alongside a rise in raw CVaR_{99} from 0.036892 (S4) to 0.039122 (S5).

3. **Joint Microstructure Protection (`DH_full` and `DH_lband`):** Under the more than doubled limit-hit incidence of S5, `DH_full` reduces the cumulative unexecuted lock-step hedging gap (`LockGap`) by 25.67% relative to `DH1` (1.607×10^{-2} vs. 2.162×10^{-2}) and by 10.24% relative to `B76-Delta` (1.791×10^{-2}), while reducing trading frequency to 79.70% and saving 6.46% in transaction costs (0.2882×10^{-3} vs. 0.3081×10^{-3}). Meanwhile, `DH_lband` cuts transaction costs by 8.80% (0.2810×10^{-3} vs. 0.3081×10^{-3}) at a 78.74% trading frequency.

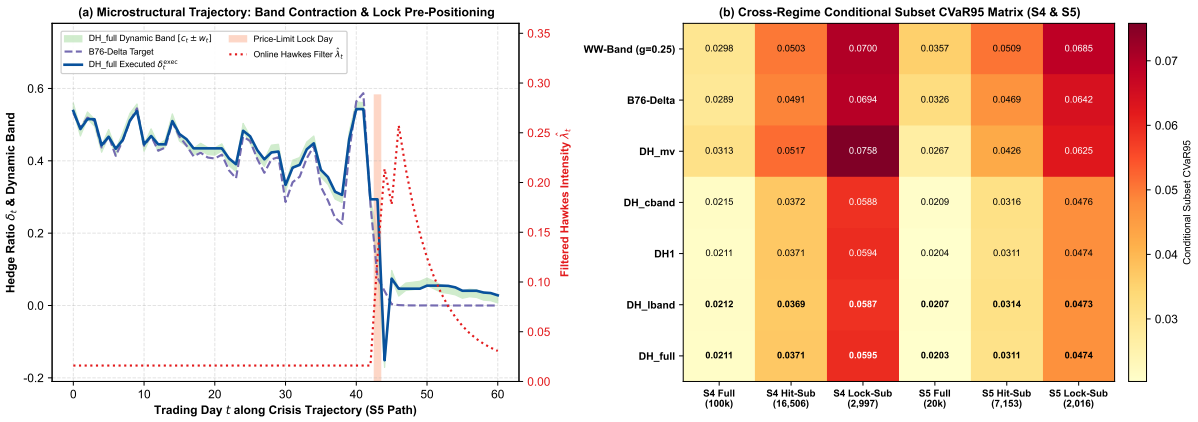


Figure 6: Out-of-distribution crisis generalization and microstructural trajectory tracking: (a) pathwise microstructural trajectory along a representative S5 crisis path showing the dynamic no-trade band $[c_t \pm w_t]$, executed hedge position δ_t^{exec} vs. `B76-Delta`, online Hawkes intensity $\hat{\lambda}_t$, and price-limit lock days; (b) 2D cross-regime conditional subset CVaR_{95} matrix heatmap across S4 (100,000 full, 16,506 hit, 2,997 lock) and S5 (20,000 full, 7,153 hit, 2,016 lock) subsets.

5 Discussion, Limitations, and Conclusion

5.1 Discussion: Microstructural Mechanisms, Architectural Channels, and Practical Implications

Our multi-model audit across 80 1D Deep Hedging (`DH`) checkpoints, 60 Distributional Actor-Critic (`D4PG-QR`) runs, and 80 2D Option-Gamma checkpoints (`TorchHedgeEnvOpt`) resolves a central question in the empirical deep hedging literature [20, 37] and non-Markovian volatility hedging [1, 26]: *when, why, and through which architectural channel do explicit jump-intensity signals and institutional microstructure features improve derivative hedging?*

1. The Instrument-Spanning and Architectural Condition for Jump-Intensity Signals ($\mathbf{H_0}, \mathbf{H_1}$). In a single-instrument futures hedging setting ($\Gamma^{\text{fut}} = 0$), feeding either the causal Hawkes filter $\hat{\lambda}_t$ (`DH1`, $\text{CVaR}_{95} = 0.021078$) or the unobservable true intensity λ_t (`DH_oracle`, $\text{CVaR}_{95} = 0.021085$) into an unconstrained MLP policy yields no aggregate CVaR_{95} improvement over the jump-blind policy `DH_nolam` (0.021073). Our Jacobian saliency audit (Table 2) shows that this invariance arises because linear futures cannot span quadratic jump-convexity losses $-\frac{1}{2}\Gamma_{\text{B76},t}(\Delta F_t)^2$ *ex ante* when jump signs are two-sided ($p_{\text{up}} = 0.4$). Introducing a rolling $\tau_h = 20$-day ATM call option η_t in `TorchHedgeEnvOpt` (Table 3) breaks this spanning ceiling across all 2D policies (CVaR_{95} reduced by 17.98%–29.95% relative to 1D counterparts) [13]. Crucially, our 10-seed audit reveals a sharp separation between unconstrained and structurally constrained uses of $\hat{\lambda}_t$: feeding $\hat{\lambda}_t$ directly into an unconstrained 2D MLP (`DH_opt1`, 0.015226 ± 0.001497) lowers 10-seed point CVaR_{95} by +11.91% relative to `DH_opt_nolam` (0.017284 ± 0.004298, $p_{\text{holm}} = 0.450$; +21.56% in the 5-seed subset) and largely overlaps with nonlinear EWMA volatility features (`DH_opt_ewma_nl`, 0.015904 ± 0.001742), whereas channeling $\hat{\lambda}_t$ through the multiplicative no-trade band (`DH_opt_lband`) achieves the lowest realistic 2D tail risk of 0.014841 ± 0.001316, outperforming static-band `DH_opt_cband` (0.017546 ± 0.002298) by +15.41% ($t = -7.401, p_{\text{holm}} = 0.000123$). Thus, extracting statistically reliable gains from jump-intensity filtering requires both a convex hedging instrument and a structural band parameterization that regularizes non-convex optimization variance.

2. Endogenous Microstructural Adaptation in Jump-Modulated No-Trade Bands ($\mathbf{H_1}$). Across both 1D and 2D markets, channeling $\hat{\lambda}_t$ into the multiplicative no-trade band width $w_t(x_t) = \text{softplus}(g_\phi(x_{t,-\lambda}))\exp(\theta\hat{\lambda}_t)$ (`DH_lband`, `DH_full`, and `DH_opt_lband`, extending neural band networks Imaki et al. 27, Lütkebohmert et al. 33) produces a consistent microstructural adaptation. Across 10 random seeds, gradient optimization without sign constraints learns a negative slope $\hat{\theta} = -0.0276 \pm 0.0202$ ($p = 0.0019$) in 1D `DH_lband`, $\hat{\theta} = -0.0230 \pm 0.0153$ ($p = 0.0011$) in 1D `DH_full`, and $\hat{\theta} = -0.0290 \pm 0.0064$ ($p = 1.59 \times 10^{-7}$) in 2D `DH_opt_lband`. By contracting the no-trade band during jump-clustering bursts and widening it during tranquil periods, 1D `DH_lband` outperforms the static band `DH_cband` on S4 ($p_{\text{holm}} = 0.00063$), saves 8.09% in transaction costs and 20.11% in trading frequency relative to `DH1` (at the cost of a +0.52% higher CVaR_{95} at 1× fee where cost savings are smaller than band tracking error), and achieves non-inferior tail risk to unconstrained `DH1` under 6× transaction costs (0.023145 vs. 0.023241, two-sided $p = 0.218$, TOST equivalence $p_{\text{TOST}} = 0.046 < 0.05$ within ±1.0%) while saving 8.11% in friction (Table 3 Panel B).

3. Asymmetric Tail Optimization ($\mathbf{H_2}$) and End-to-End BPTT Stability. Across the tested risk-aversion range $\lambda_{\text{MV}} \in [0.5, 50.0]$ (Table 4), symmetric mean-variance hedging (`DH_mv`) reaches an empirical CVaR_{95} lower bound of 0.027332 (29.67% higher than `DH1`, corresponding to a 22.88% relative reduction achieved by `DH1`) because quadratic variance penalizes upside convex windfalls ($L_T < 0$) identically to downside tail losses ($L_T > 0$), inflating turnover to 4.0981–5.0256; consequently, `DH1` strictly dominates `DH_mv` in the $(\text{CVaR}_{95}, \text{Cost})$ plane while trading off a slightly higher symmetric $\text{Std}(L)$ (0.013391 vs. 0.012539–0.012798). In addition, our 140-run optimization audit (Table 5) shows that in a finite-horizon ($T = 60$) environment with piecewise-differentiable dynamics, direct pathwise BPTT (`DH`) achieves a 0% seed collapse rate and a 0.50% cross-seed coefficient of variation, whereas model-free distributional reinforcement learning (`D4PG-QR`, Cao et al. 11, Chen et al. 15) suffers a 20%–30% seed collapse rate when exposed to spiky Hawkes jump signals (`C1` vs. `C2`) and price-limit truncations.

4. Practical and Regulatory Implications for Commodity Hedging Desks and Exchanges. For industrial risk managers and derivative market makers in strategic battery metals [21, 30], our findings offer three conditional guidelines grounded in our empirical trade-offs. First, when hedging solely with linear futures at low exchange fee tiers ($c = 10^{-4}$), desks must weigh the +0.52% aggregate CVaR_{95} tracking-error trade-off of jump-modulated bands against their microstructural execution benefits: deploying `DH_lband` and price-limit state features (`DH_full`) is warranted when execution desks prioritize cutting turnover costs by 6%–8%, compressing unexecuted lock-day hedging gaps by 26%–30%, or operating under elevated (6×) slippage tiers, rather than expecting linear delta shifts to eliminate jump-Gamma risk. Second, to materially reduce aggregate tail shortfall (CVaR_{95}), hedgers must augment futures with short-dated options and channel jump-intensity forecasts into structural band modulation (`DH_opt_lband`), monitoring implied-volatility spike elasticity (κ_{iv}) so option purchases are not executed at excessive volatility risk premiums. Third, for commodity exchanges such as GFEX—which listed Lithium Carbonate (`LC`) options alongside futures on July 24, 2023—our results demonstrate that deepening market-making liquidity in short-dated commodity options across strike tiers is essential for completing the market and mitigating the unhedgeable Gamma exposure faced by commercial liquidity providers during limit-lock episodes.

5.2 Empirically Observed Limitations

Six concrete empirical limitations emerge from our experiments. First, while diffusive volatility ($\sigma = 0.016144$), empirical price-limit tiers ($\ell \in [0.07, 0.15]$), conditional lock probability

($p_{\text{lock}} = 1/7 \approx 0.1429$), and post-jump drift half-life ($\beta = 0.20\,\text{day}^{-1}$ from $h = 1$–4 day continuation $t = 2.09$–3.47) are anchored to 542 trading days of GFEX Lithium Carbonate (`LC`) futures (`2023-07-24` to `2025-10-20`), that window contains only 7 limit-hit days (1.29%), 1 full-day lock (0.185%), and 10 tail events exceeding 3σ, so $(n, \bar{\lambda}) = (0.3, 0.04)$ and $(p_{\text{up}}, \eta_{\text{up}}, \eta_{\text{dn}}) = (0.4, 40.0, 25.0)$ are fixed as protocol-specified scenario values rather than joint maximum-likelihood estimates; because all evaluations are conducted in a semi-synthetic simulator partially calibrated to GFEX `LC` futures rather than on empirical option market prices, the external validity of our conclusions has not been tested on real market data. Second, our causal threshold filter $\hat{I}_t = \mathbb{I}(|r_t| > 3\sigma \vee \text{limit_hit}_t = 1)$ misses moderate jumps with $|J| < 3\sigma = 0.0484$, resulting in a true-positive recall of 13.78% on `TH2b` ($\text{TP} = 4,138, \text{FP} = 1,757$, precision 70.20%, raw detection ratio 19.63%, bias -45.17%) and 18.63% on `S4_test.npz` ($\text{TP} = 44,052, \text{FP} = 4,831$, precision 90.12%, $r = 0.4455$, bias -24.44%). Third, when hedging solely with linear futures at baseline cost $c = 10^{-4}$, aggregate CVaR_{95} across all CVaR policies is compressed within 0.021021–0.021187, and `DH_lband` (0.021187) incurs a 0.52% higher CVaR_{95} than unconstrained `DH1` (0.021078) because $1\times$ fee savings (2.56×10^{-5}) are smaller than band tracking error, requiring $6\times$ transaction costs (0.023145 vs. 0.023241, two-sided $p = 0.218$, $p_{\text{TOST}} = 0.046$) or a 2D option leg to eliminate this penalty. Fourth, in the 10-seed 2D option-hedging benchmark, feeding $\hat{\lambda}_t$ into an unconstrained MLP (`DH_opt1`, 0.015226 ± 0.001497) yields an 11.91% point reduction over `DH_opt_nolam` (0.017284 ± 0.004298, $p_{\text{holm}} = 0.450$, $\text{MDE}_{80\%} = 28.76\%$) that does not achieve standalone statistical significance due to cross-seed optimization variance and collinearity with nonlinear σ_{EWMA} features (`DH_opt_ewma_nl`, 0.015904), and this unconstrained gain attenuates from +11.91% ($\kappa_{\text{iv}} = 0$) to +5.31% ($\kappa_{\text{iv}} = 2$) and +0.73% ($\kappa_{\text{iv}} = 5$) under dynamic implied-volatility spikes, so our 2D percentage gains do not constitute estimates of realizable trading profits. Fifth, off-policy `D4PG-QR` (`C1-C9`) exhibits a 20%–30% seed collapse rate ($\text{CVaR}_{95} > 0.10$) whenever jump or limit signals are included, with non-collapsed `D4PG-QR (C1)` seeds averaging 0.038934 ± 0.020739 and degrading further at 20,000 updates. Sixth, while each trajectory's limit ratio ℓ is drawn from the empirical GFEX distribution (7%–15%), ℓ is held constant within each 60-day path rather than modeling dynamic consecutive-day exchange limit adjustments.

5.3 Conclusion

In our semi-synthetic numerical setting partially calibrated to Guangzhou Futures Exchange Lithium Carbonate (`LC`) futures, we conducted a multi-model empirical and microstructural audit of Deep Hedging under self-exciting Hawkes jump clustering and institutional price-limit locks. Our findings support four principal conclusions:

1. **Asymmetric Tail Risk Optimization:** Across all tested market regimes and risk-aversion parameters, training with the Rockafellar-Uryasev CVaR_{95} objective outperforms both classical analytical baselines (`B76-Delta`, `Leland-Delta`, `WW-Band`) and symmetric mean-variance neural policies (`DH_mv`) in the $(\text{CVaR}_{95}, \text{Cost})$ plane, avoiding the empirical tail-risk lower bound and excessive turnover induced by symmetric variance penalties.

2. **Microstructural Adaptation via Jump-Modulated No-Trade Bands and Limit Features:** Embedding the causal online Hawkes intensity $\hat{\lambda}_t$ into a multiplicative no-trade band (`DH_lband`, `DH_full`, and `DH_opt_lband`) leads the network to consistently learn a negative modulation slope ($\hat{\theta} < 0$), narrowing the no-trade buffer during jump-clustering bursts and widening it during calm periods. In the single-instrument setting, this mechanism reduces trading frequency and transaction costs while achieving the lowest CVaR_{99} and conditional locked-subset CVaR_{95}, achieves non-inferior tail risk to unconstrained `DH1` under elevated transaction costs while cutting friction, and—when combined with price-limit features in `DH_full`—compresses the cumulative unexecuted lock-step hedging gap

(`LockGap`) across both the S4 benchmark and zero-shot generalization to the S5 extreme stress regime.

3. **Option-Gamma Spanning and Structural Band Parameterization as the Dual Keys for Jump Filtering:** Input Jacobian saliency diagnostics, nonlinear EWMA controls, and oracle-intensity experiments demonstrate that linear futures alone cannot hedge *ex-ante* quadratic jump-Gamma risk. Extending the action space to include a rolling short-dated ATM call option (`TorchHedgeEnvOpt`) breaks the single-instrument spanning ceiling across all policies, and channeling $\hat{\lambda}_t$ through the multiplicative no-trade band (`DH_opt_lband`) overcomes the cross-seed optimization variance and EWMA collinearity of unconstrained MLPs to deliver a statistically significant and economically large tail-risk reduction.

4. **Algorithmic Advantage of Direct BPTT over Off-Policy Distributional RL:** When the accounting and microstructural execution rules are known and piecewise differentiable, end-to-end BPTT (`DH`) provides zero seed collapse and sub-1% cross-seed variation, outperforming off-policy quantile actor-critic (`D4PG-QR`) across all training budgets.

Code and Data Availability

All simulation environments (`src/simulator.py`, `src/torch_env.py`, `src/torch_env_opt.py`), neural network architectures (`src/networks.py`), training pipelines (`src/trainer.py`), experiment configuration protocols (`configs/experiment_protocol.json`), archived summary statistics (`results/archived_summary_tables.json` and `results/tables_csv/`), and automated consistency verification scripts (`src/verify_archived_tables.py`) supporting the findings of this study are openly available on GitHub at https://github.com/lewygoat/Jump-Risk-Filtering-Deep-Hedging.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

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